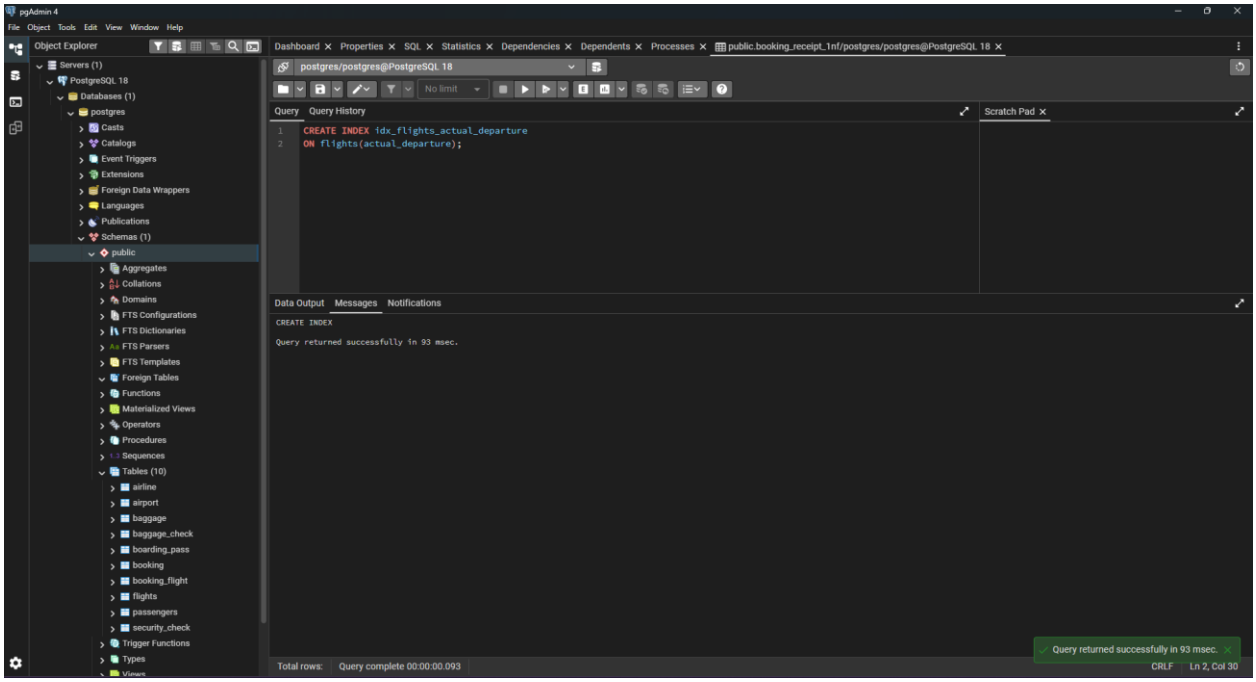
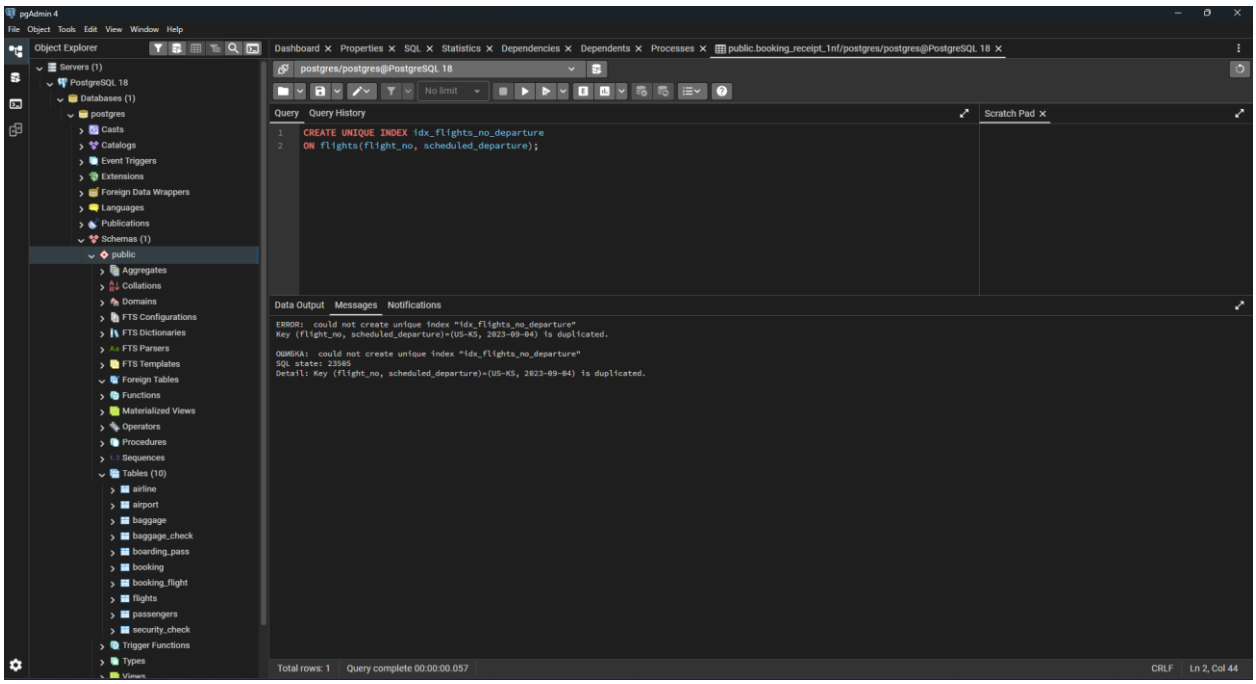
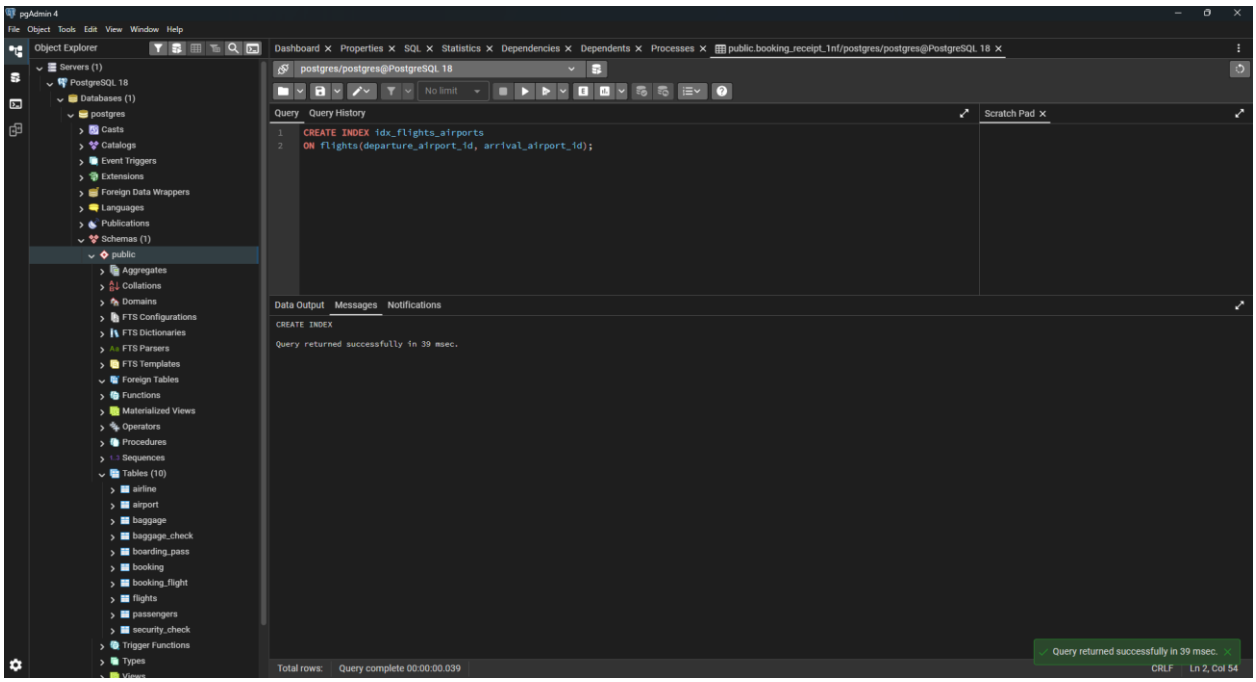
1. Create an index on the actual_departure column in the flights table.



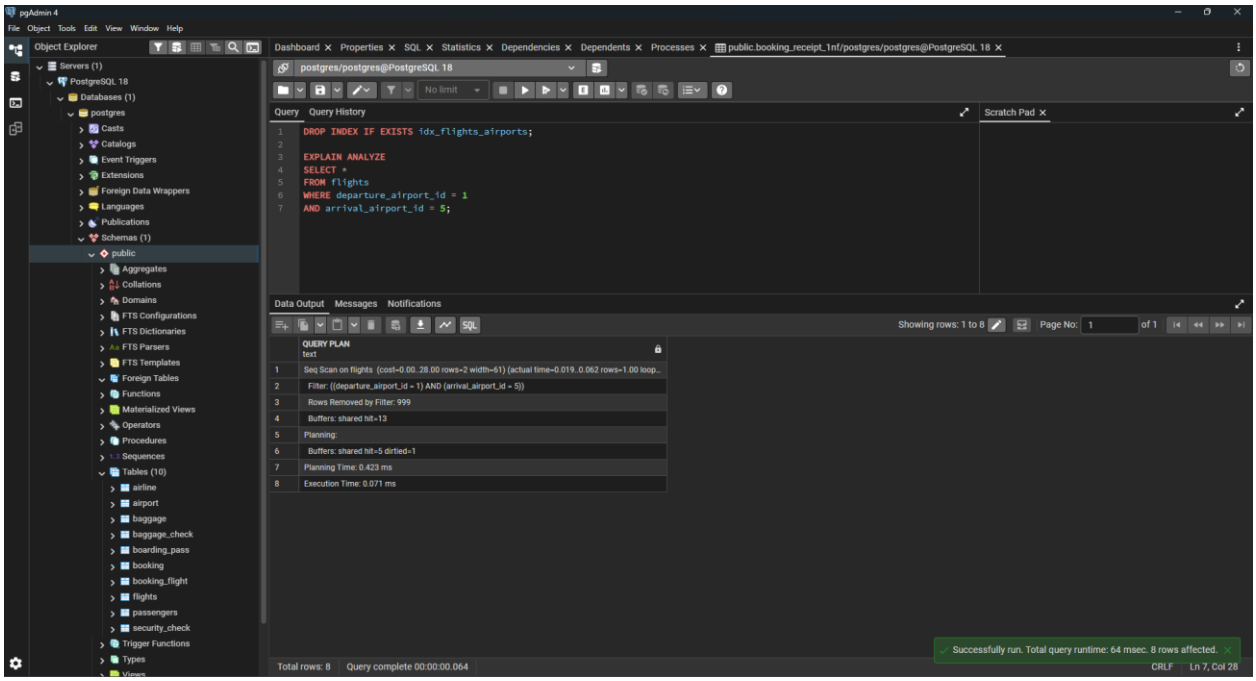
2. Create a unique index to ensure flight_no and scheduled_departure combinations are unique.

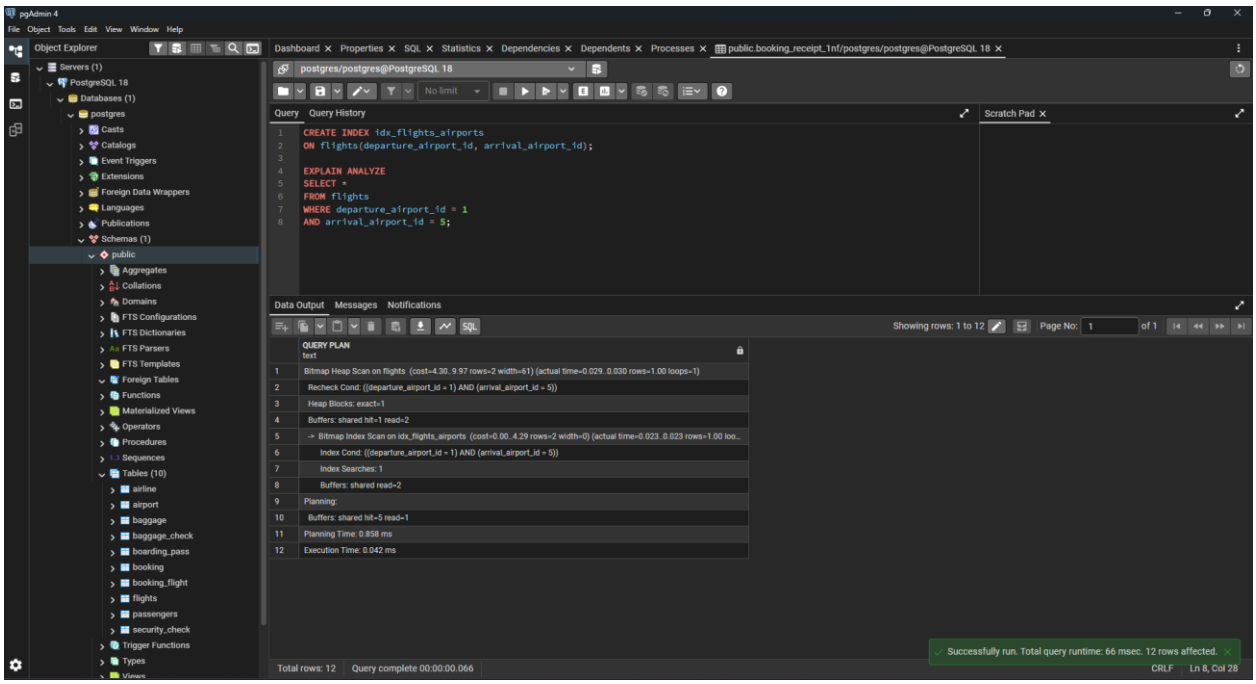


3. Create a composite index on the departure_airport_id and arrival_airport_id columns.

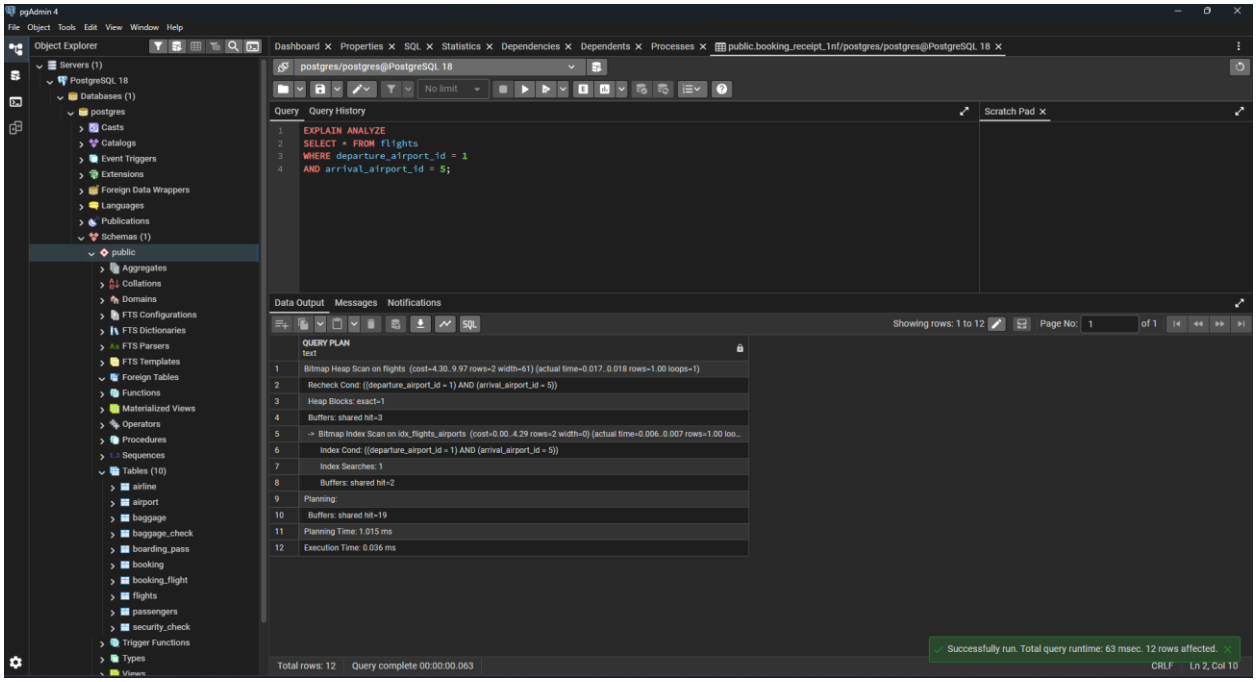


4. Evaluate the difference in query performance with and without indexes. Measure performance differences.

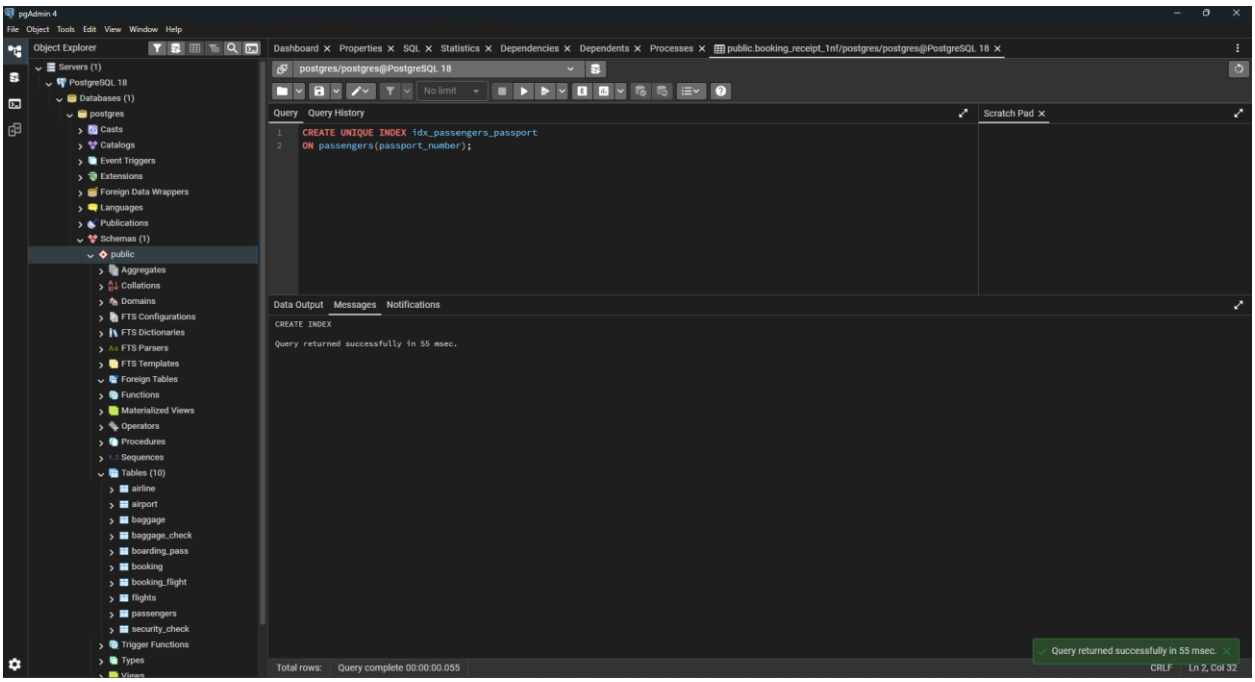


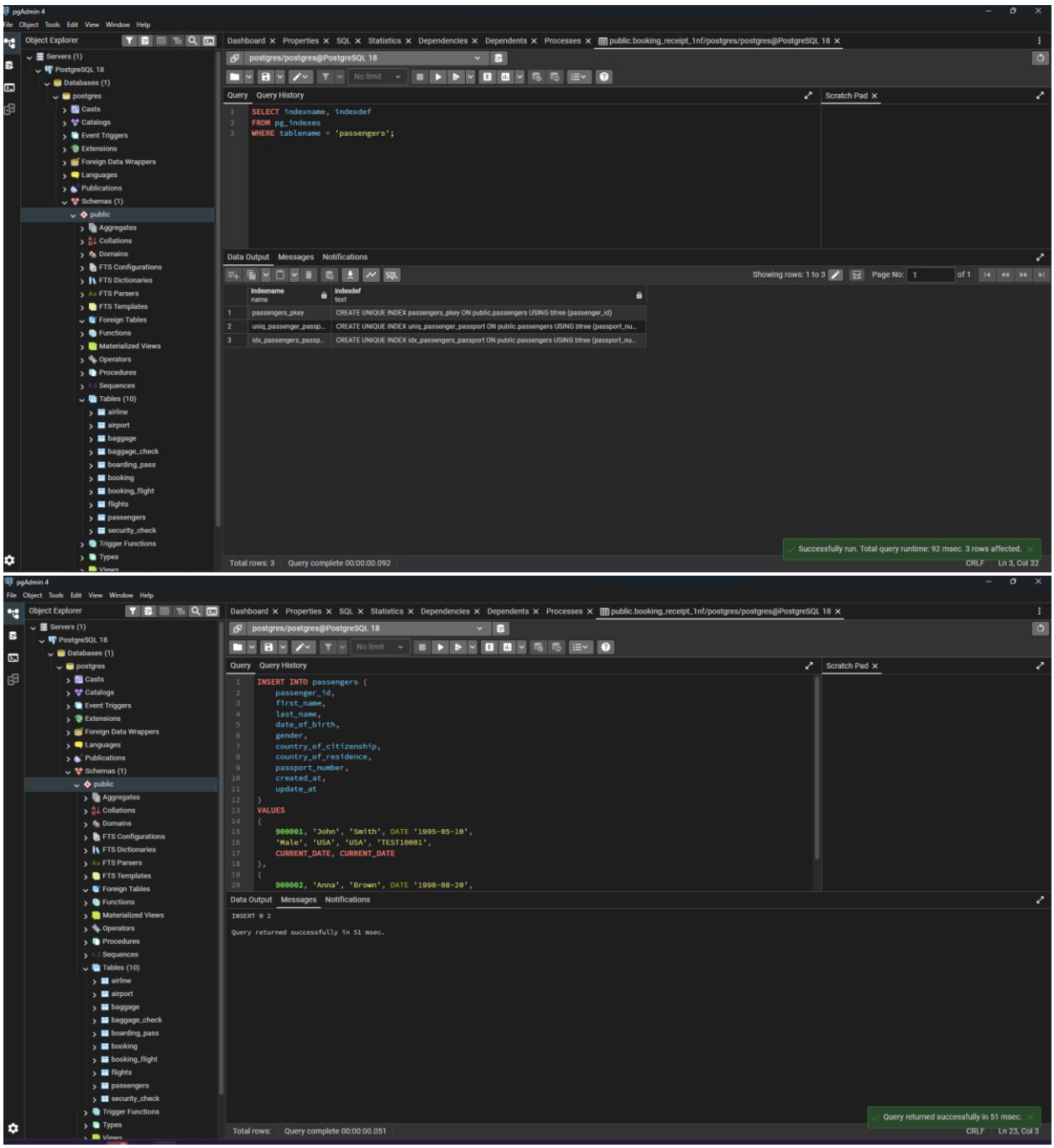


5. Use EXPLAIN ANALYZE to check index usage in a query filtering by departure_airport and arrival_airport.



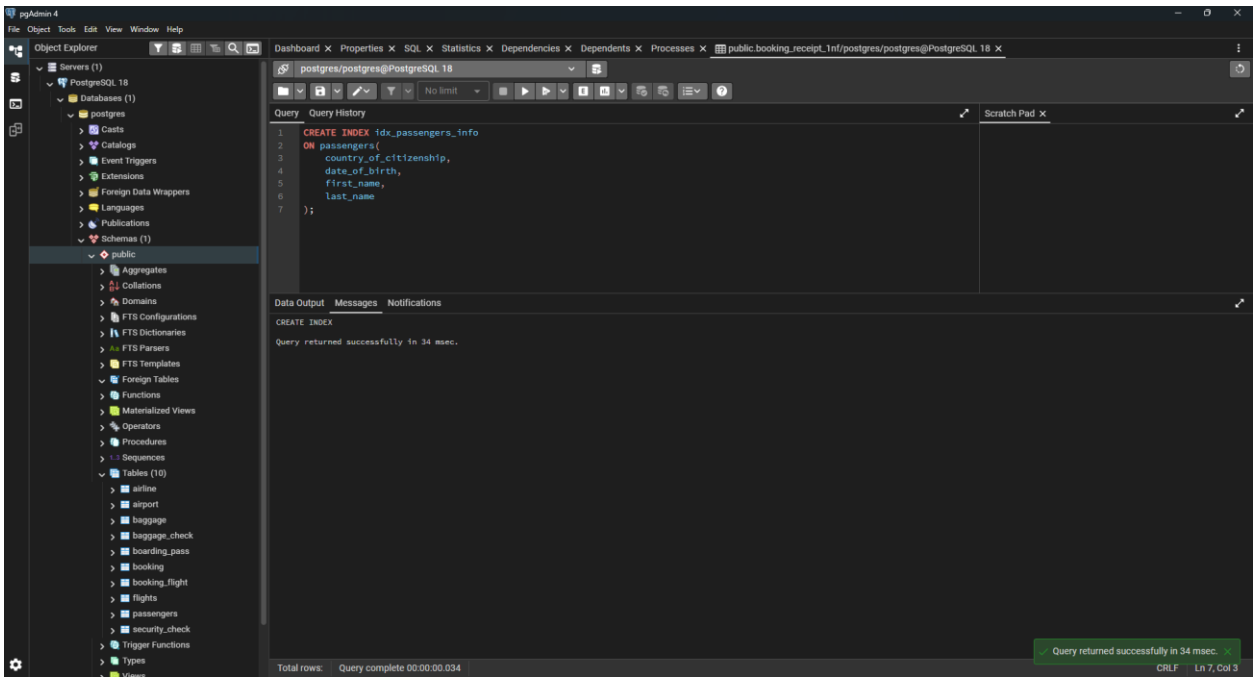
6. Create a unique index for the passport_number of the Passengers table. Check if the index was created or not. Insert into the table two new passengers. Explain in your own words what is going on in the output?

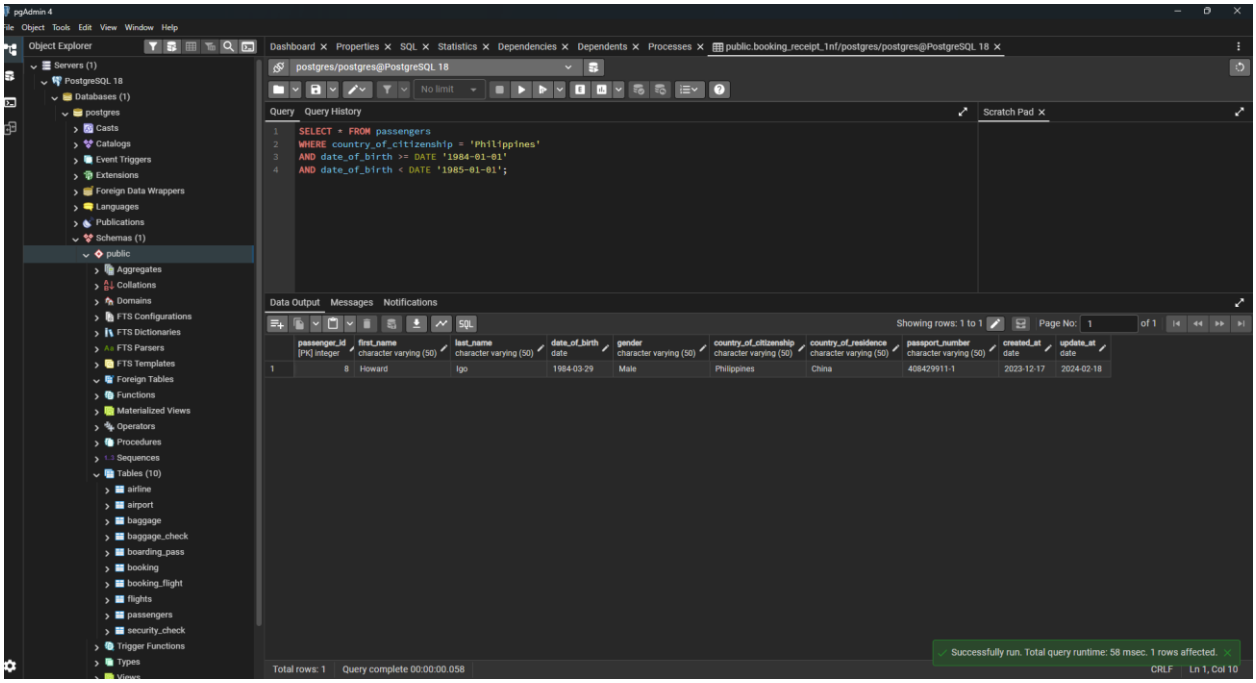

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public.booking_receipt_1nf/postgres/postgres@PostgreSQL 18
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1 CREATE UNIQUE INDEX idx_passengers_passport
2 ON passengers(passport_number);
Data Output
Messages
Notifications
CREATE INDEX
Query returned successfully in 55 msec.
Query returned successfully in 55 msec.
Total rows:
Query complete 00:00:00.055
CRLF
Ln 2, Col 32

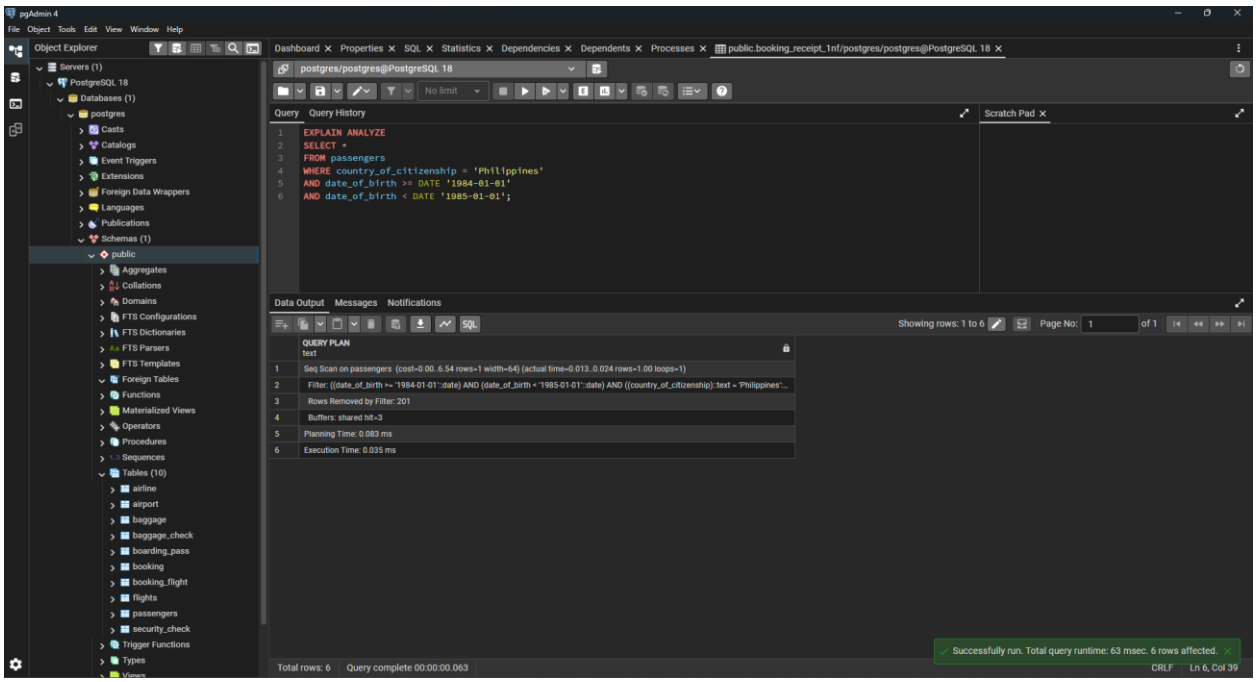


The unique index checks passport numbers and prevents duplicate values. When we insert a passenger with an already existing passport number, PostgreSQL rejects the insertion.

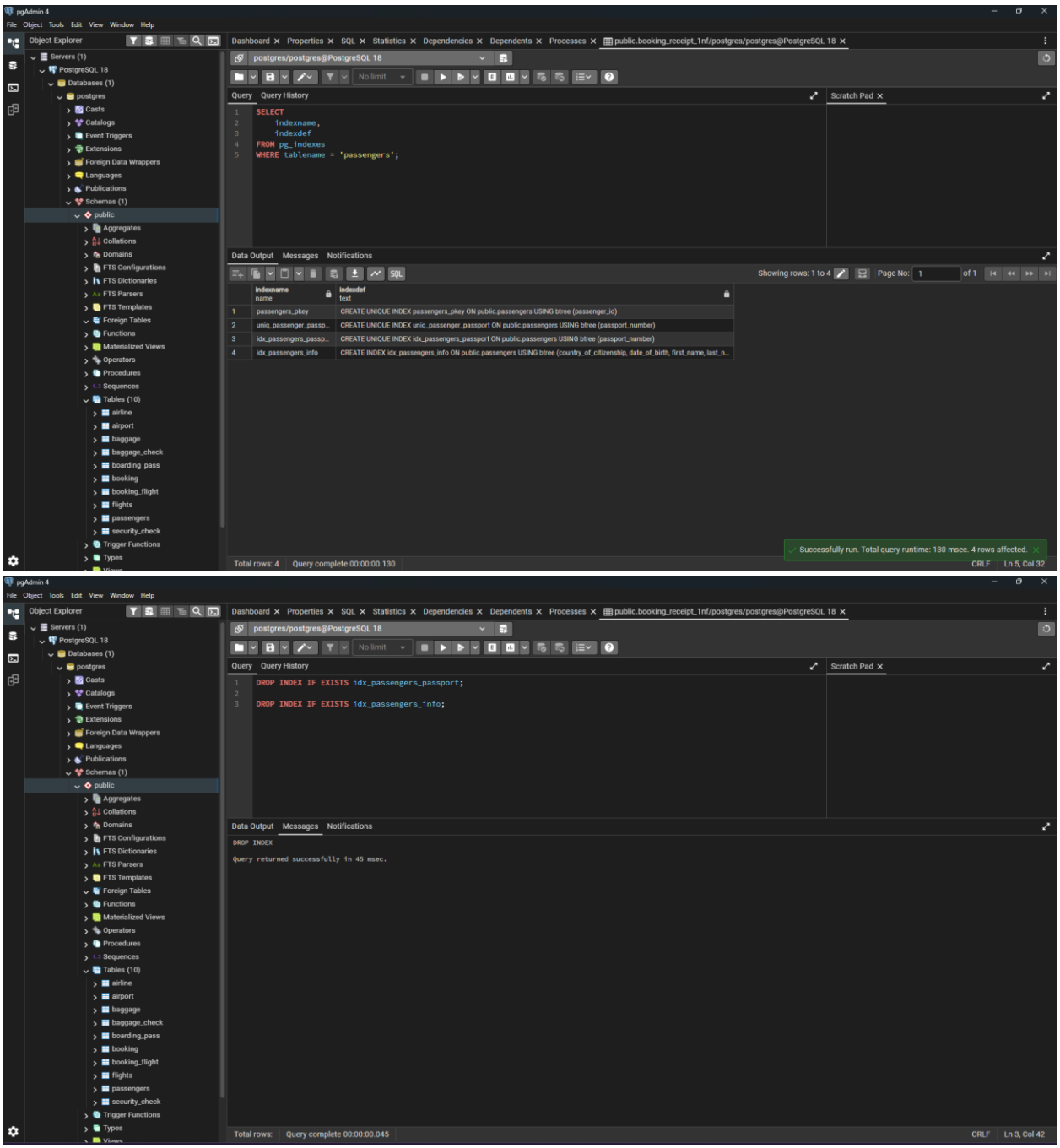
7. Create an index for the Passengers table. Use for that first name, last name, date of birth and country of citizenship. Then, write a SQL query to find a passenger who was born in Philippines and was born in 1984 and check if the query uses indexes or not. Give an explanation of the results.


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public.booking_receipt_1nf/postgres/postgres@PostgreSQL 18
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
CREATE INDEX idx_passengers_info
ON passengers(
country_of_citizenship,
date_of_birth,
first_name,
last_name
);
Data Output
Messages
Notifications
CREATE INDEX
Query returned successfully in 34 msec.
Query returned successfully in 34 msec.
Total rows:
Query complete 00:00:00.034
CRLF
Ln 7, Col 3

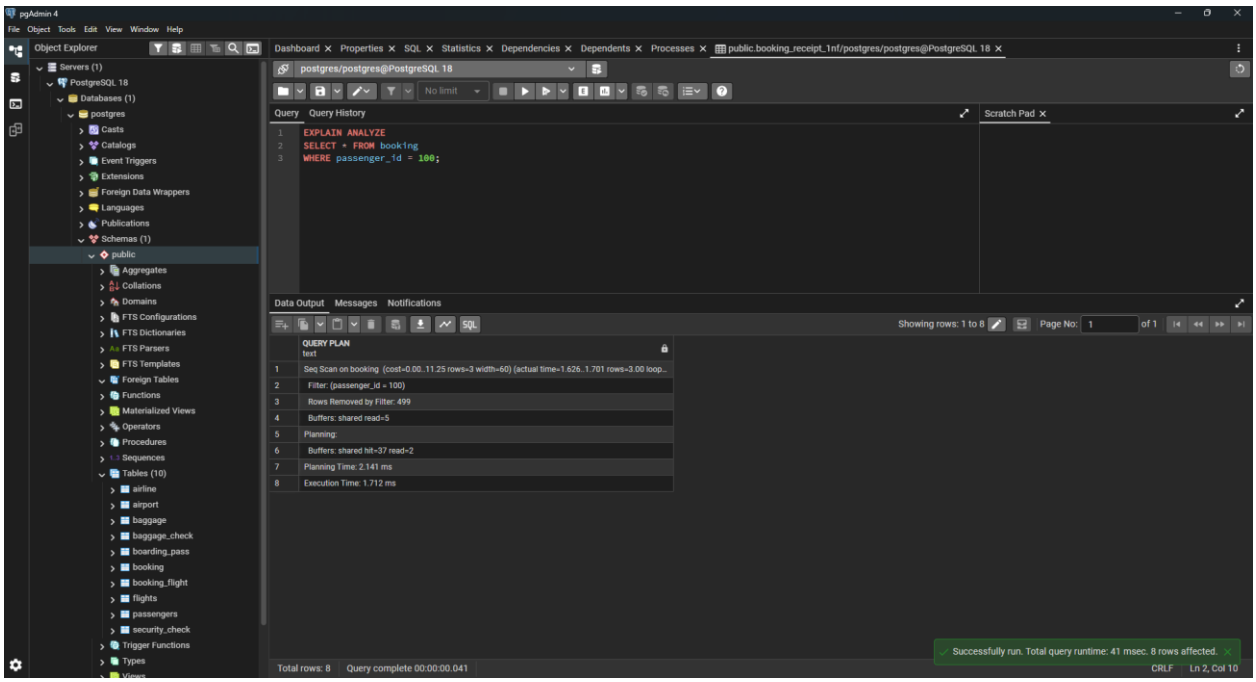

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Dependents
Processes
public.booking_receipt_1nf/postgres/postgres@PostgreSQL 18
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
SELECT * FROM passengers
WHERE country_of_citizenship = 'Phil!ppines'
AND date_of_birth >= DATE '1984-01-01'
AND date_of_birth < DATE '1985-01-01';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1
Page No: 1
of 1
passenger_id [PK] integer
first_name character varying (50)
last_name character varying (50)
date_of_birth date
gender character varying (50)
country_of_citizenship character varying (50)
country_of_residence character varying (50)
passport_number character varying (50)
created_at date
update_at date
1 8 Howard Igo 1984-03-29 Male Philippines China 408429911-1 2023-12-17 2024-02-18
Successfully run. Total query runtime: 58 msec. 1 rows affected.
Total rows: 1
Query complete 00:00:00.058
CRLF
Ln 1, Col 10

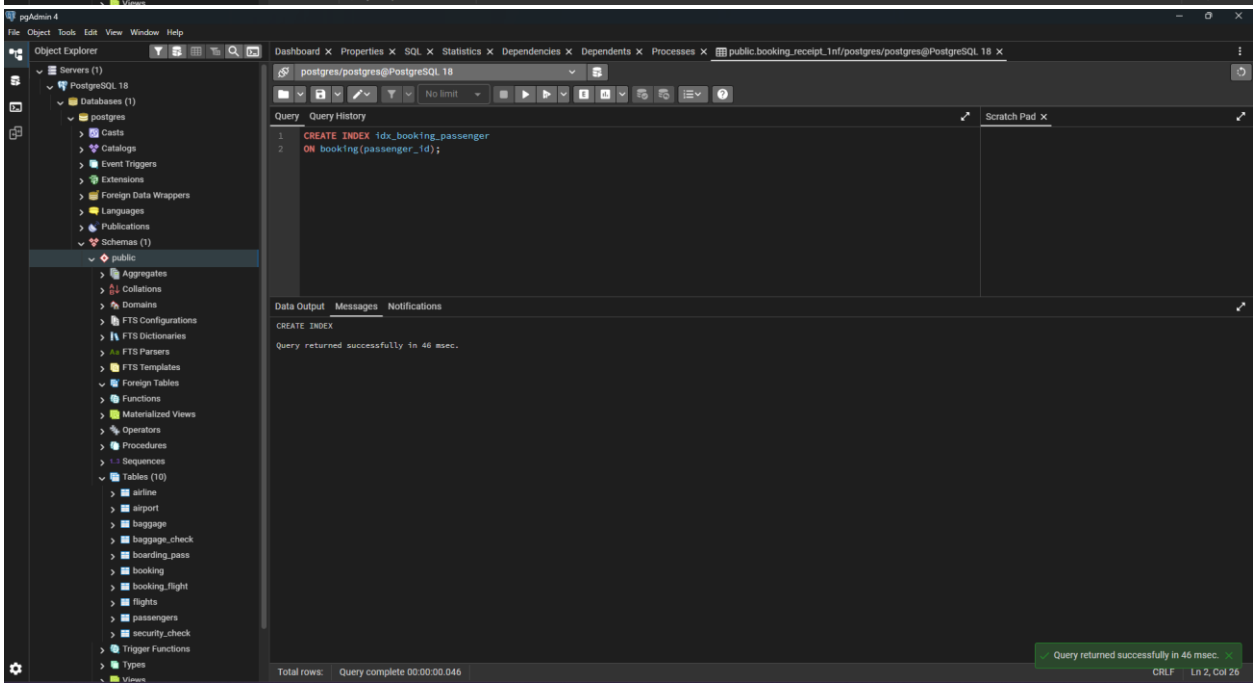


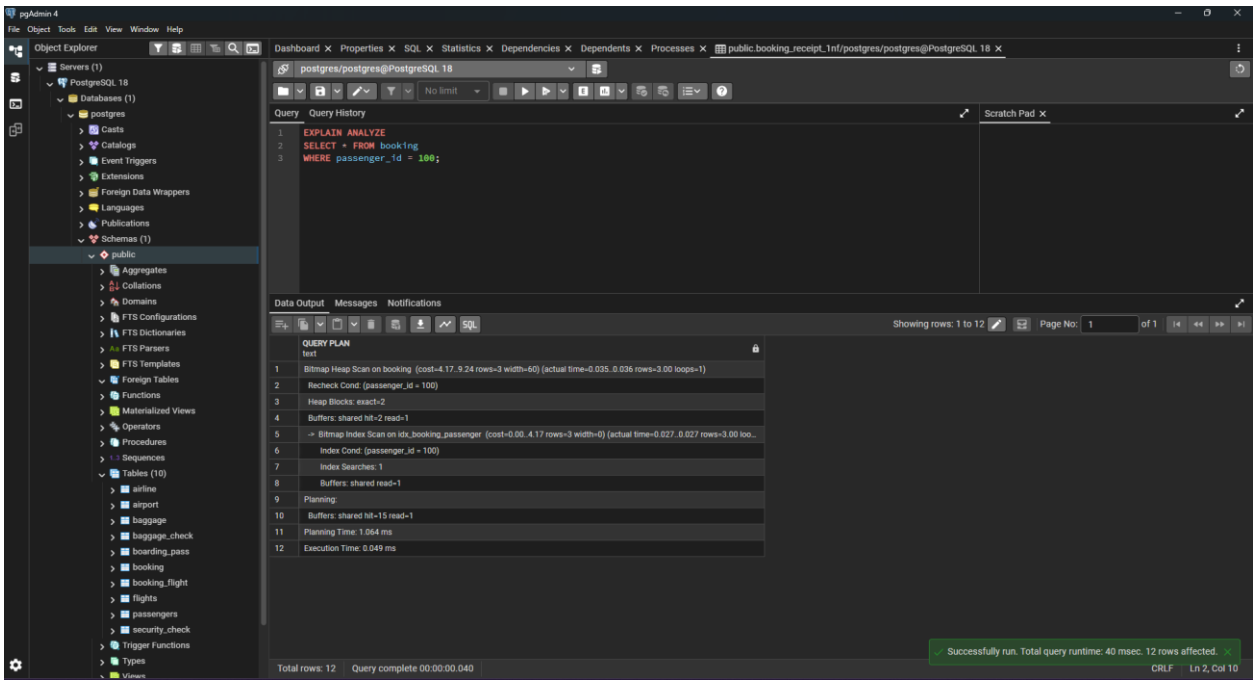
8. Write a SQL query to list indexes for table Passengers. After, delete the created indexes.



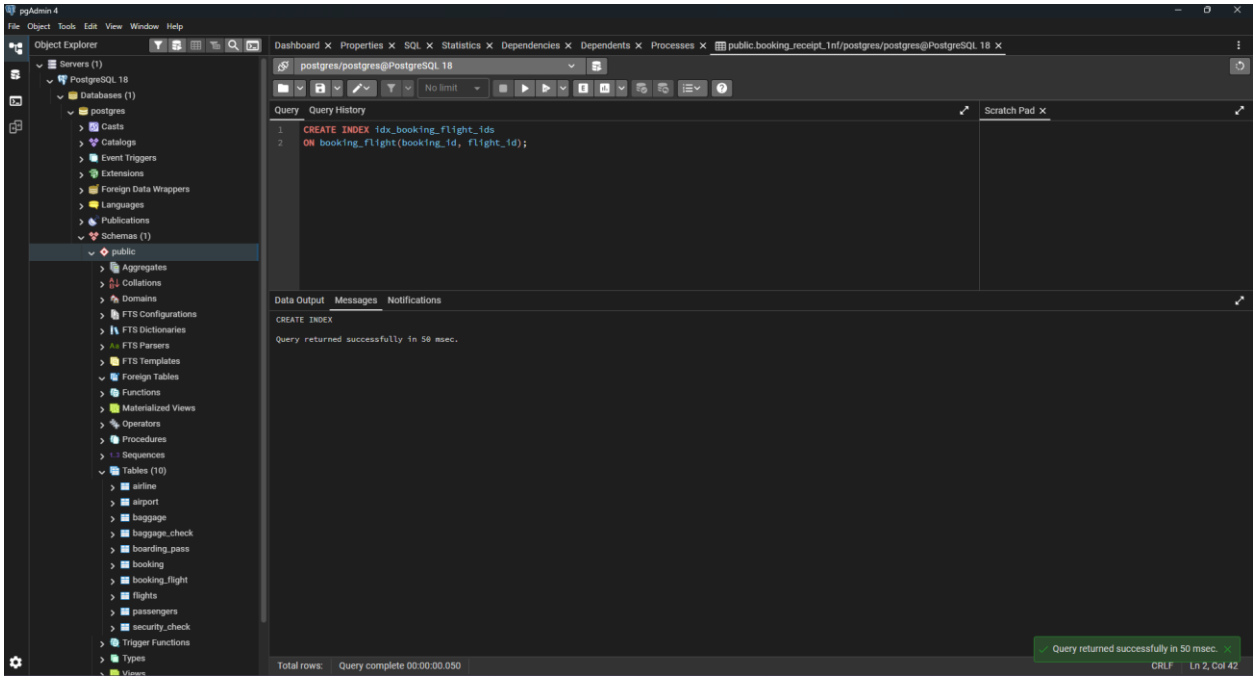
9. Create an index on the passenger_id column of the booking table. Using EXPLAIN ANALYZE, compare the execution plan and execution time of a query that selects all bookings of one passenger (for example, passenger_id = 100) before and after creating the index. Explain what changed in the plan.

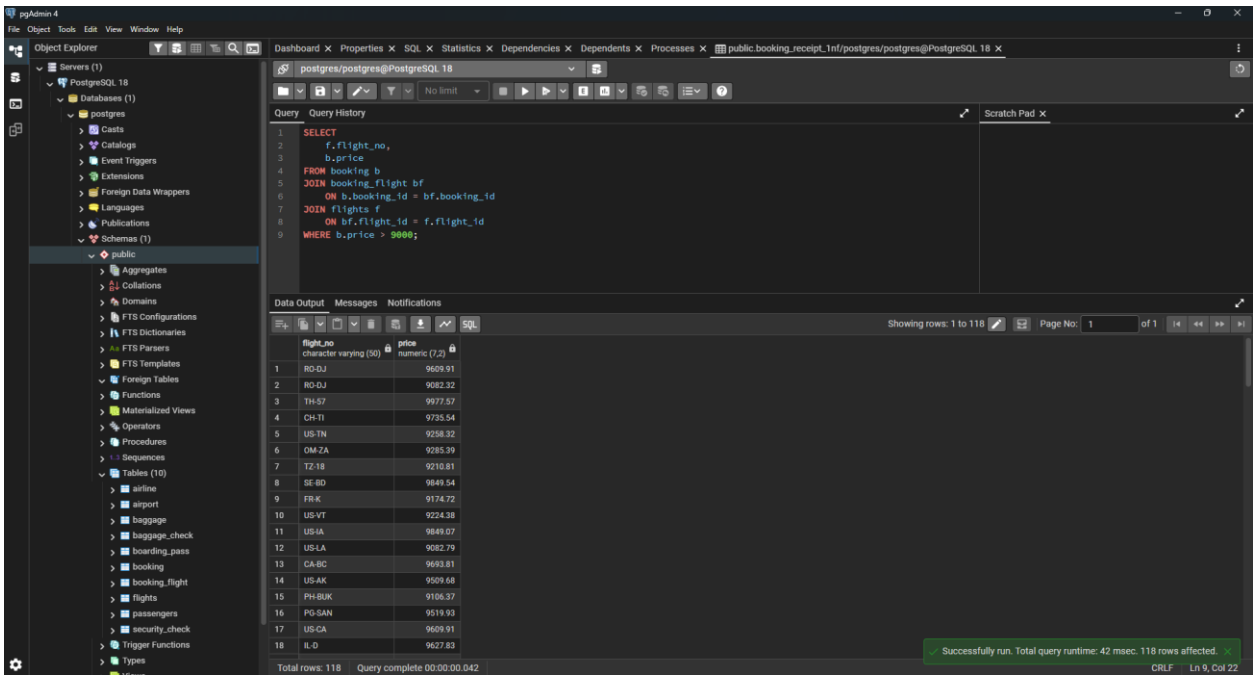

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Dependents
Processes
public.booking_receipt_1nf/postgres/postgres@PostgreSQL 18
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
1 EXPLAIN ANALYZE
2 SELECT * FROM booking
3 WHERE passenger_id = 100;
Scratch Pad
Data Output
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SQL
Showing rows: 1 to 8
Page No: 1 of 1
QUERY PLAN
text
1 Seq Scan on booking (cost=0.00..11.25 rows=3 width=60) (actual time=1.626..1.701 rows=3.00 loop...
2 Filter: (passenger_id = 100)
3 Rows Removed by Filter: 499
4 Buffers: shared read=5
5 Planning:
6 Buffers: shared hit=37 read=2
7 Planning Time: 2.141 ms
8 Execution Time: 1.712 ms
Successfully run. Total query runtime: 41 msec. 8 rows affected.
Total rows: 8
Query complete 00:00:00.041
CRLF
Ln 2, Col 10

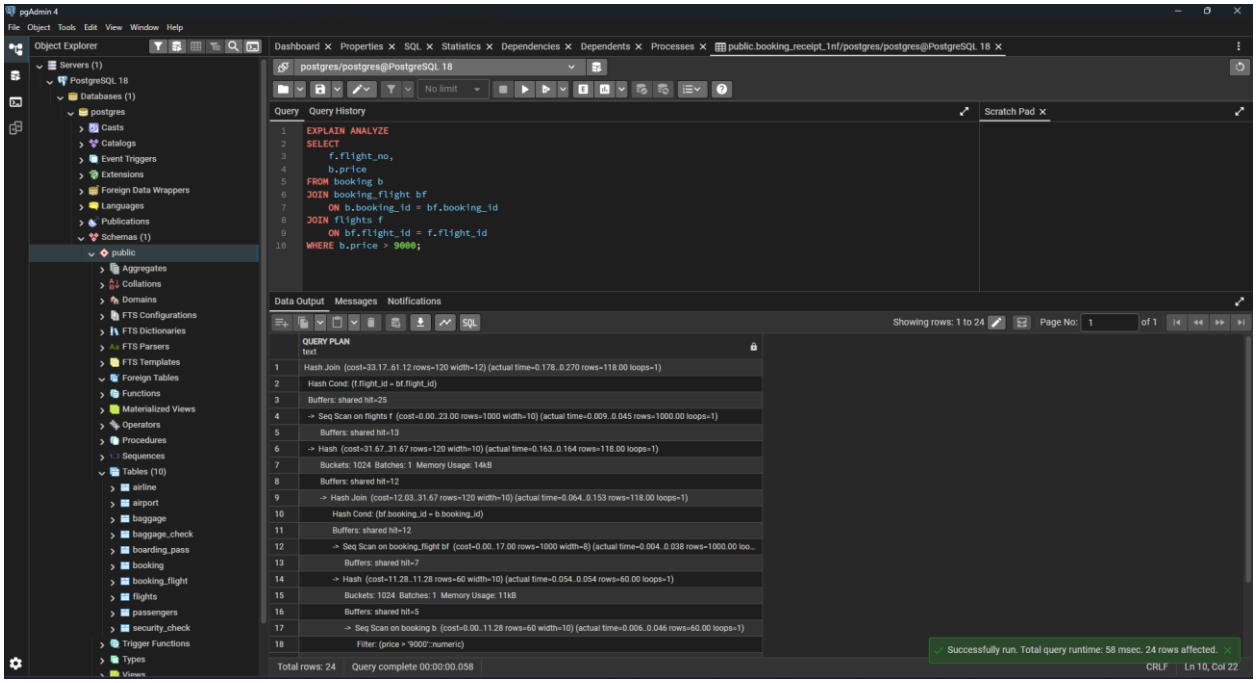

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Processes
public.booking_receipt_1nf/postgres/postgres@PostgreSQL 18
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
1 CREATE INDEX idx_booking_passenger
2 ON booking(passenger_id);
Scratch Pad
Data Output
Messages
Notifications
CREATE INDEX
Query returned successfully in 46 msec.
Query returned successfully in 46 msec.
Total rows:
Query complete 00:00:00.046
CRLF
Ln 2, Col 26



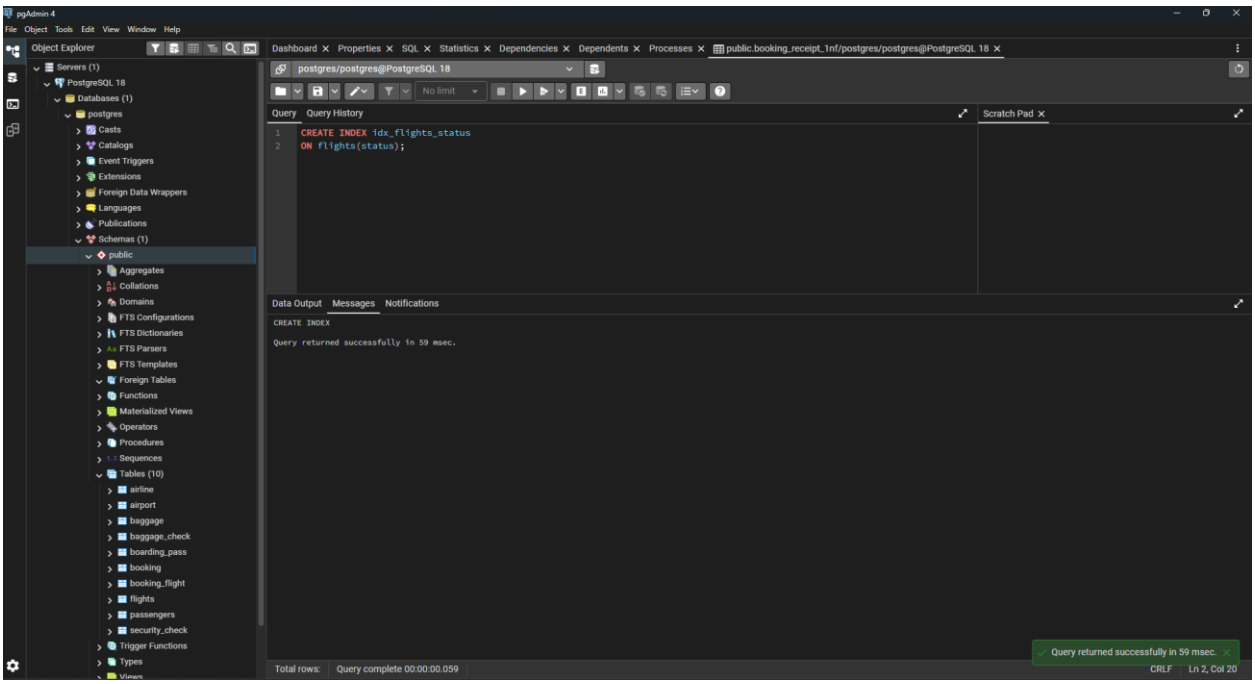
10. Create a composite index on the booking_flight table using the booking_id and flight_id columns. Write a query that joins booking, booking_flight and flights and returns the flight_no and price for bookings with a price greater than 9000. Check with EXPLAIN ANALYZE which indexes are used in each step of the join.

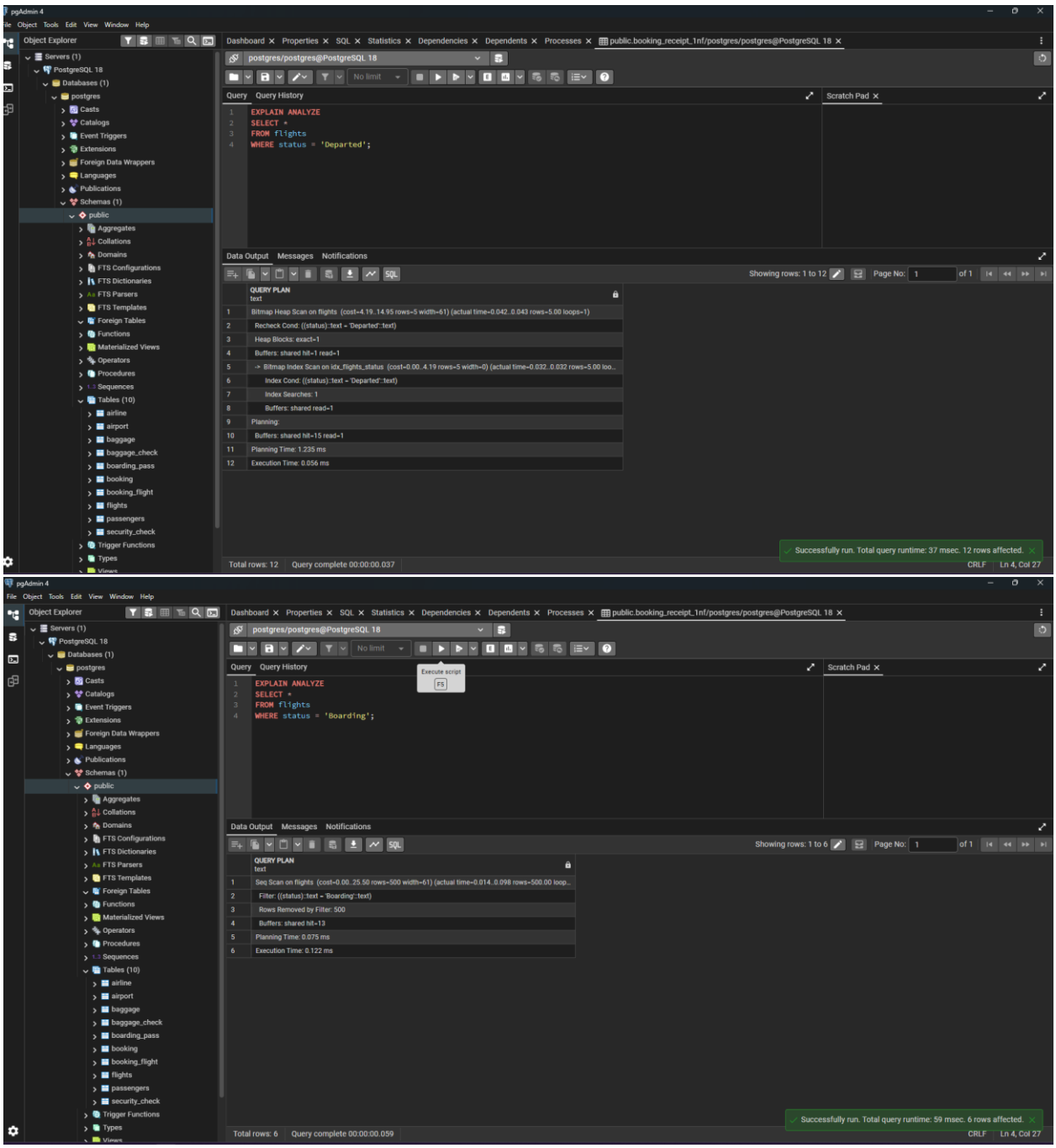




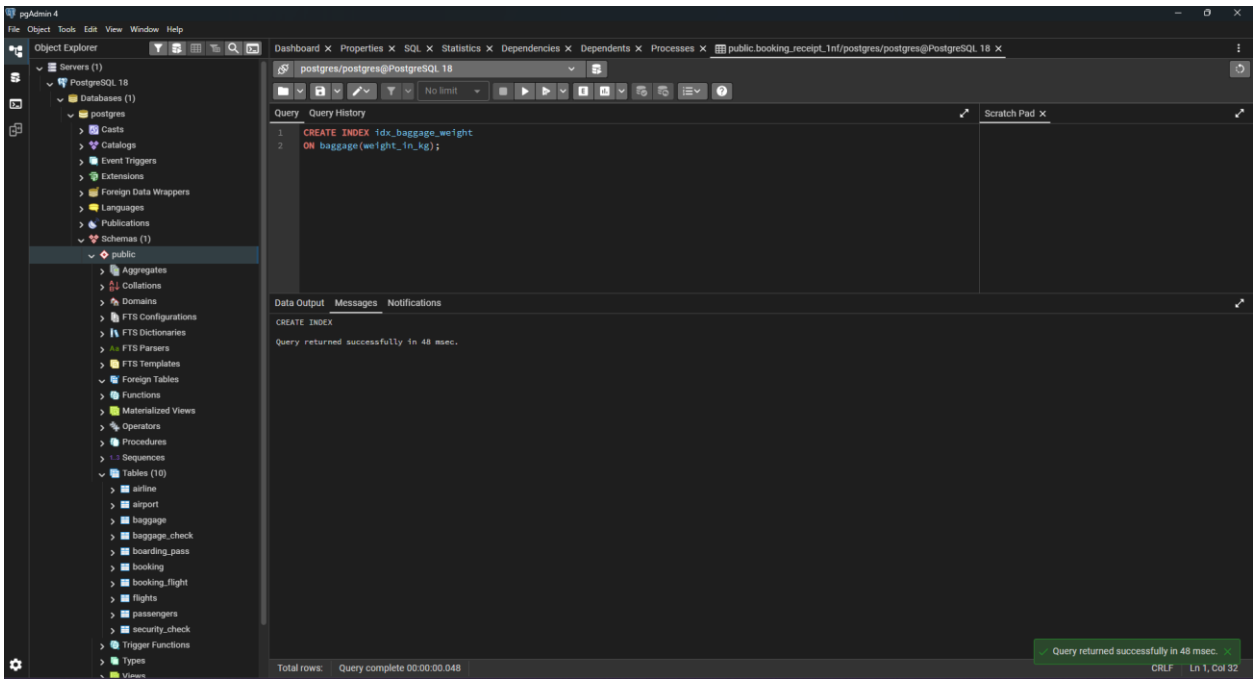


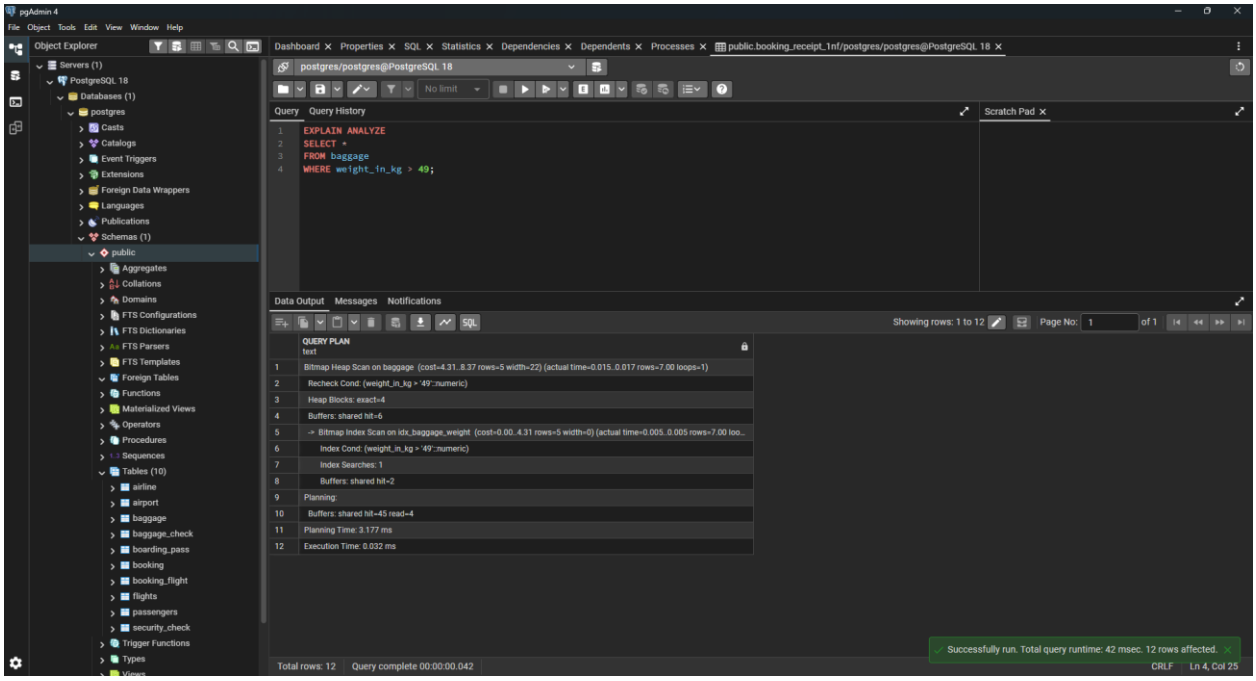
11. Create an index on the status column of the flights table. Run EXPLAIN ANALYZE for the query with status = 'Departed' and for the query with status = 'Boarding'. Compare the plans and explain, in your own words, why the optimizer may ignore the index for one of the values (hint: selectivity).

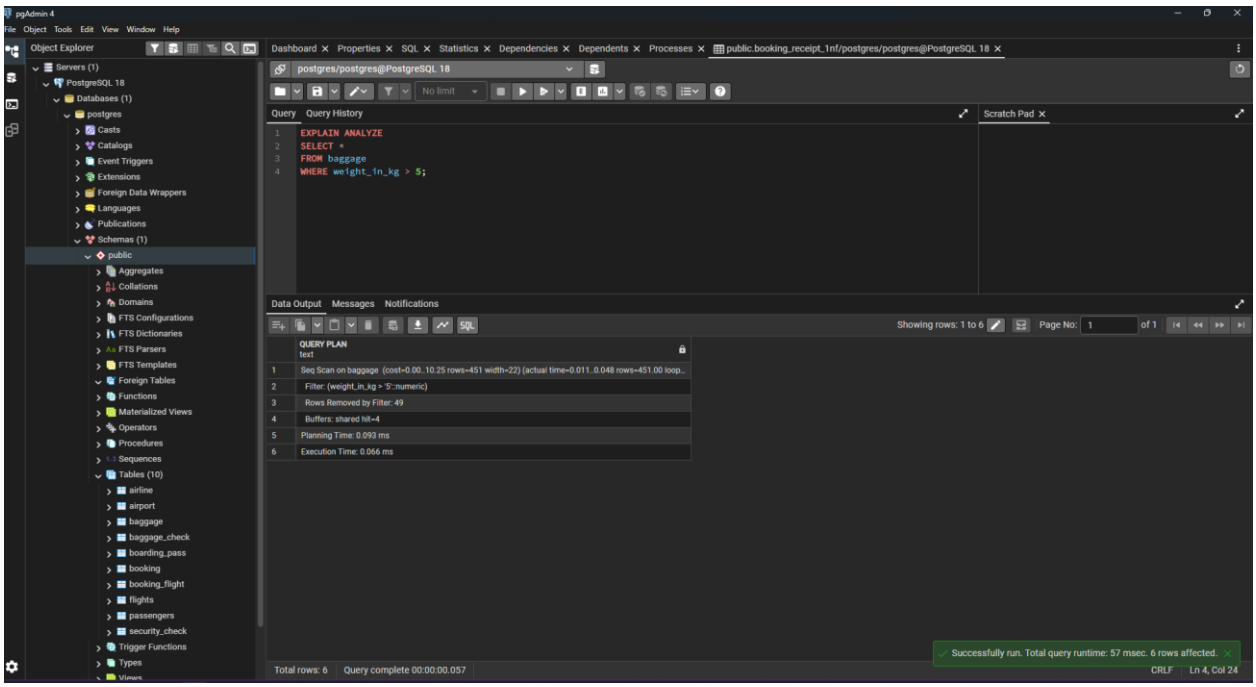

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public.booking_receipt_1nf/postgres/postgres@PostgreSQL 18
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
1 CREATE INDEX idx_flights_status
2 ON flights(status);
Scratch Pad
Data Output
Messages
Notifications
CREATE INDEX
Query returned successfully in 59 msec.
Total rows:
Query complete 00:00:00.059
Query returned successfully in 59 msec.
CRLF
Ln 2, Col 20



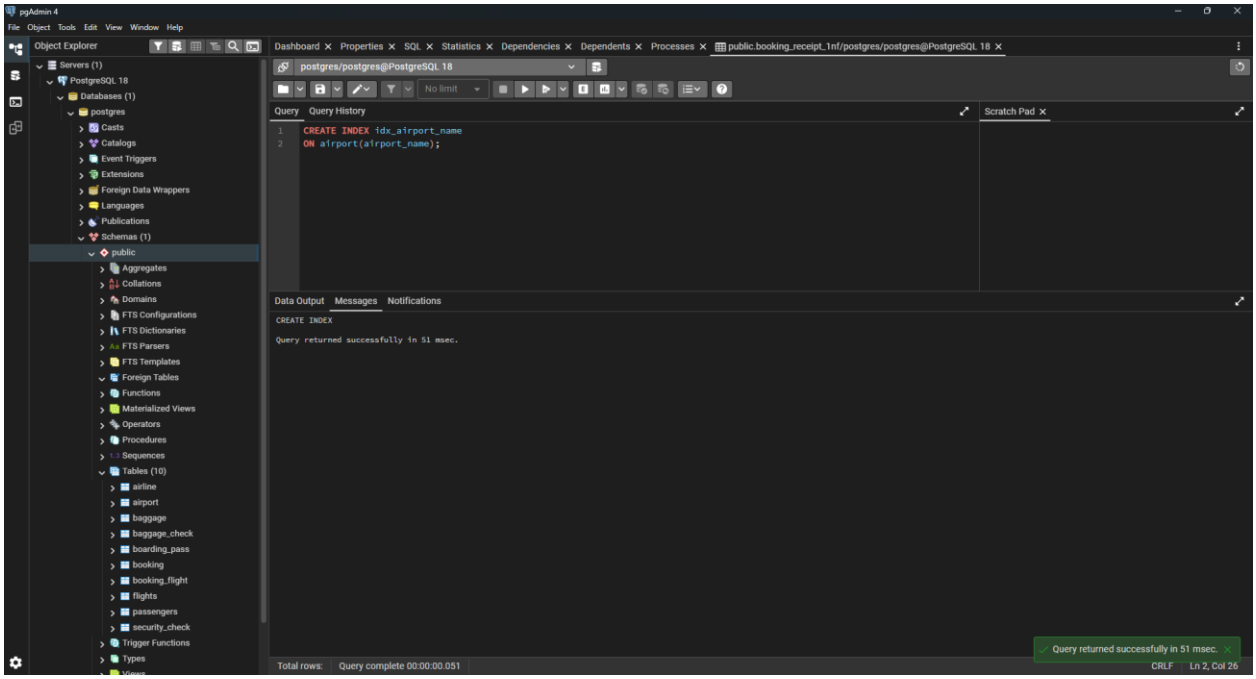
12. Create an index on the weight_in_kg column of the baggage table. Compare the plans of two range queries: weight_in_kg > 49 (a small part of the table) and weight_in_kg > 5 (almost the whole table). Explain the difference in index usage.

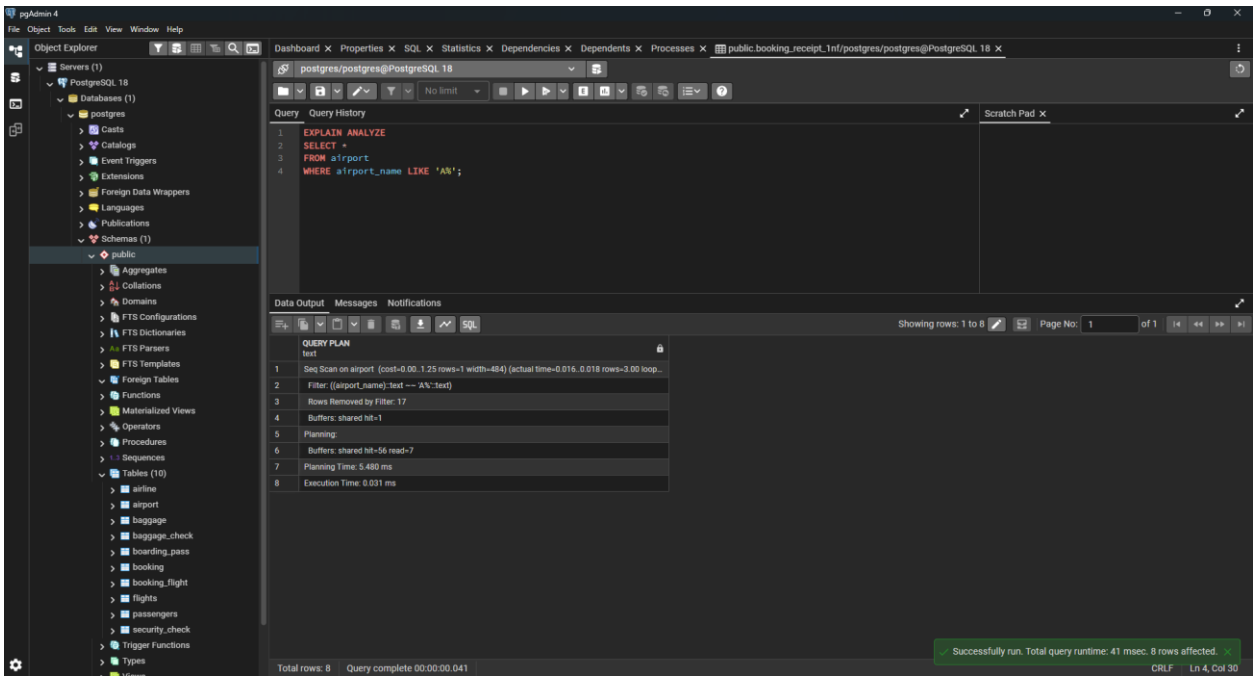

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public.booking_receipt_1nf/postgres/postgres@PostgreSQL 18
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
CREATE INDEX idx_baggage_weight
ON baggage(weight_in_kg);
Data Output
Messages
Notifications
CREATE INDEX
Query returned successfully in 48 msec.
Query returned successfully in 48 msec.
Total rows:
Query complete 00:00:00.048
CRLF
Ln 1, Col 32

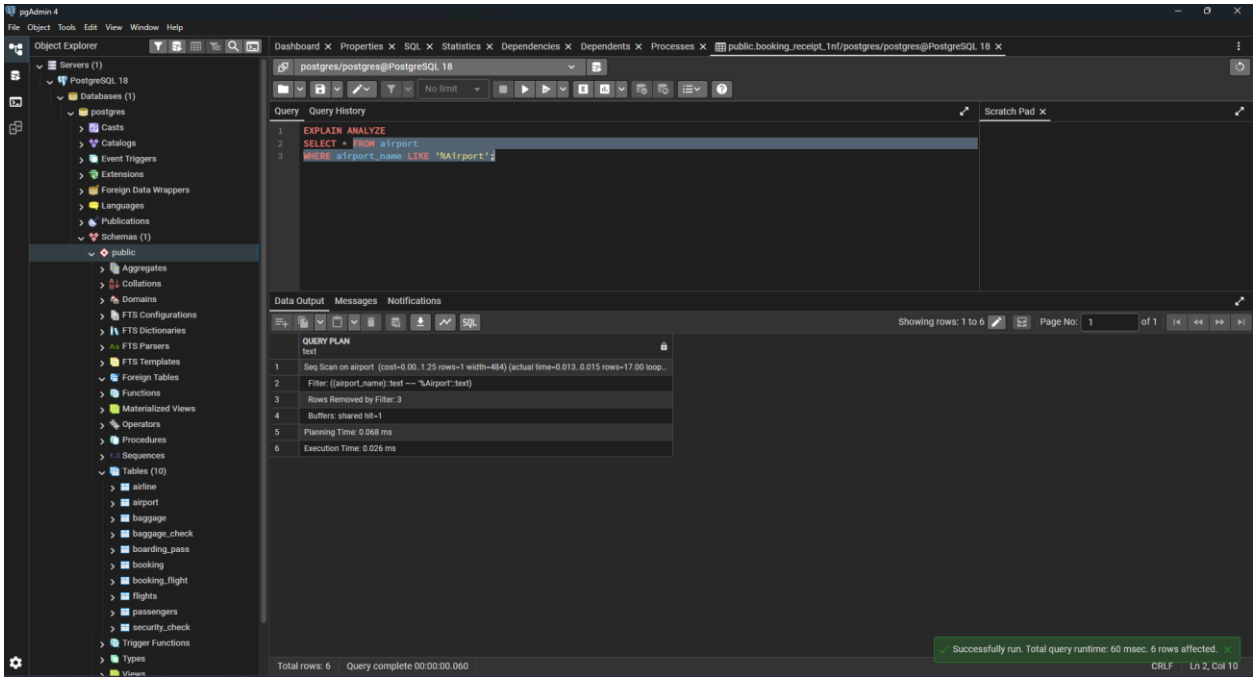

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public.booking_receipt_1nf/postgres/postgres@PostgreSQL 18
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
EXPLAIN ANALYZE
SELECT *
FROM baggage
WHERE weight_in_kg > 49;
Data Output
Messages
Notifications
Showing rows: 1 to 12
Page No: 1 of 1
QUERY PLAN
text
1 Bitmap Heap Scan on baggage (cost=4.31..8.37 rows=5 width=22) (actual time=0.015..0.017 rows=7.00 loops=1)
2 Recheck Cond: (weight_in_kg > '49'::numeric)
3 Heap Blocks: exact=4
4 Buffers: shared hit=6
5 -> Bitmap Index Scan on idx_baggage_weight (cost=0.00..4.31 rows=5 width=0) (actual time=0.005..0.005 rows=7.00 loo...
6 Index Cond: (weight_in_kg > '49'::numeric)
7 Index Searches: 1
8 Buffers: shared hit=2
9 Planning:
10 Buffers: shared hit=45 read=4
11 Planning Time: 3.177 ms
12 Execution Time: 0.032 ms
Successfully run. Total query runtime: 42 msec. 12 rows affected.
Total rows: 12
Query complete 00:00:00.042
CRLF
Ln 4, Col 25



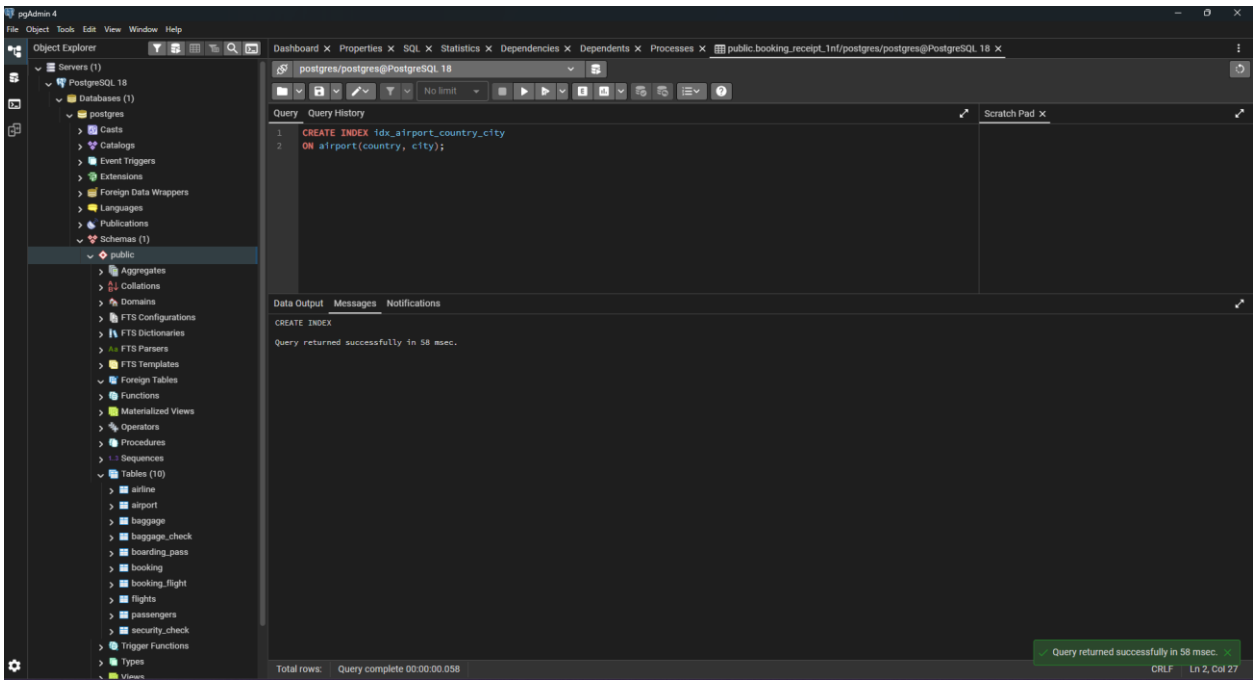
13. Create an index on the airport_name column of the airport table. Check with EXPLAIN ANALYZE whether it is used by the queries LIKE 'A%' and LIKE '%Airport'. Explain why the results are different.

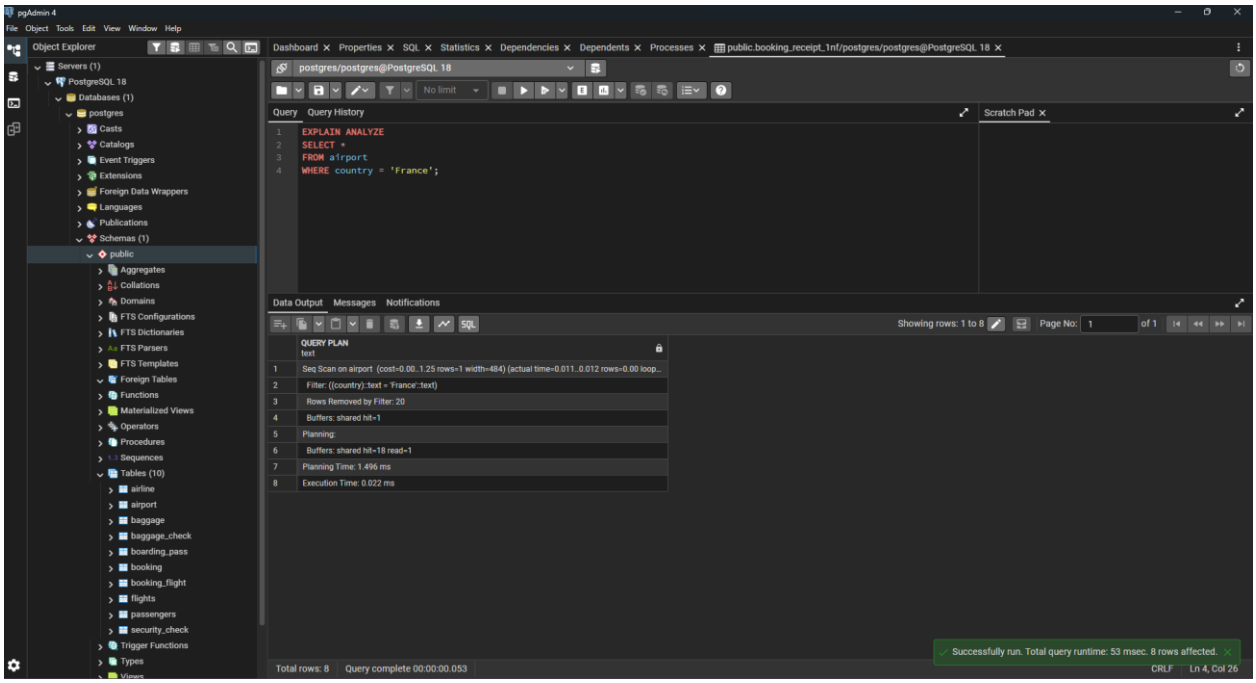


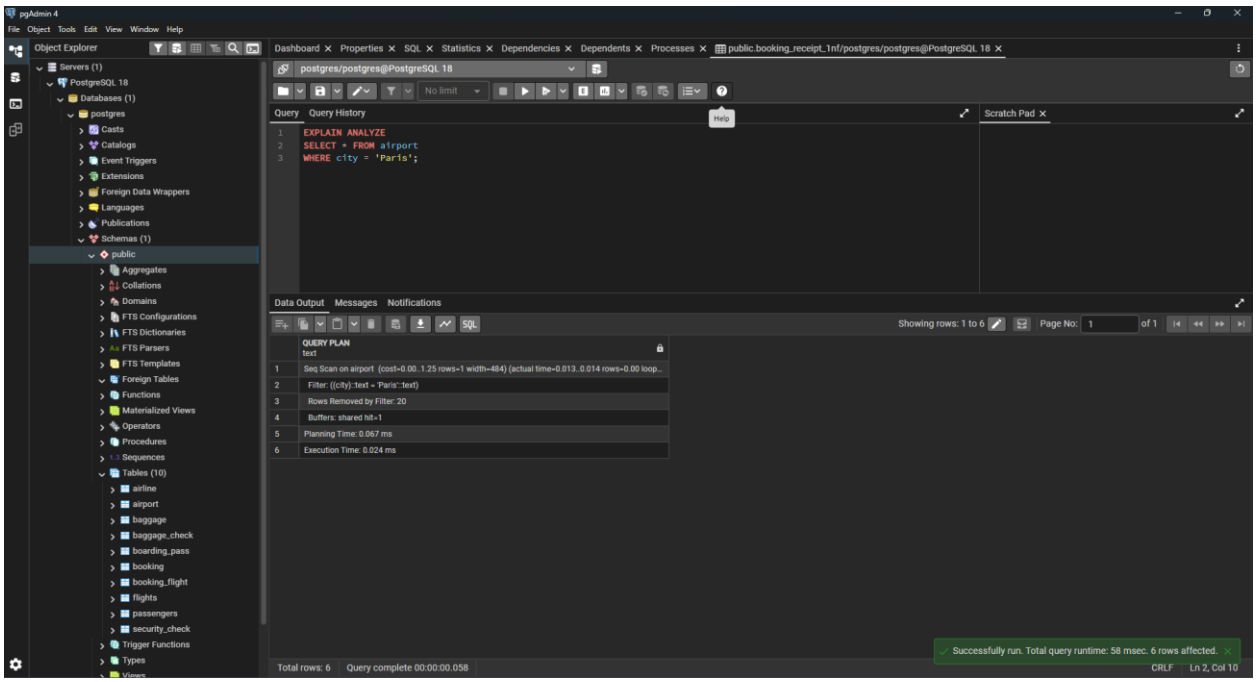


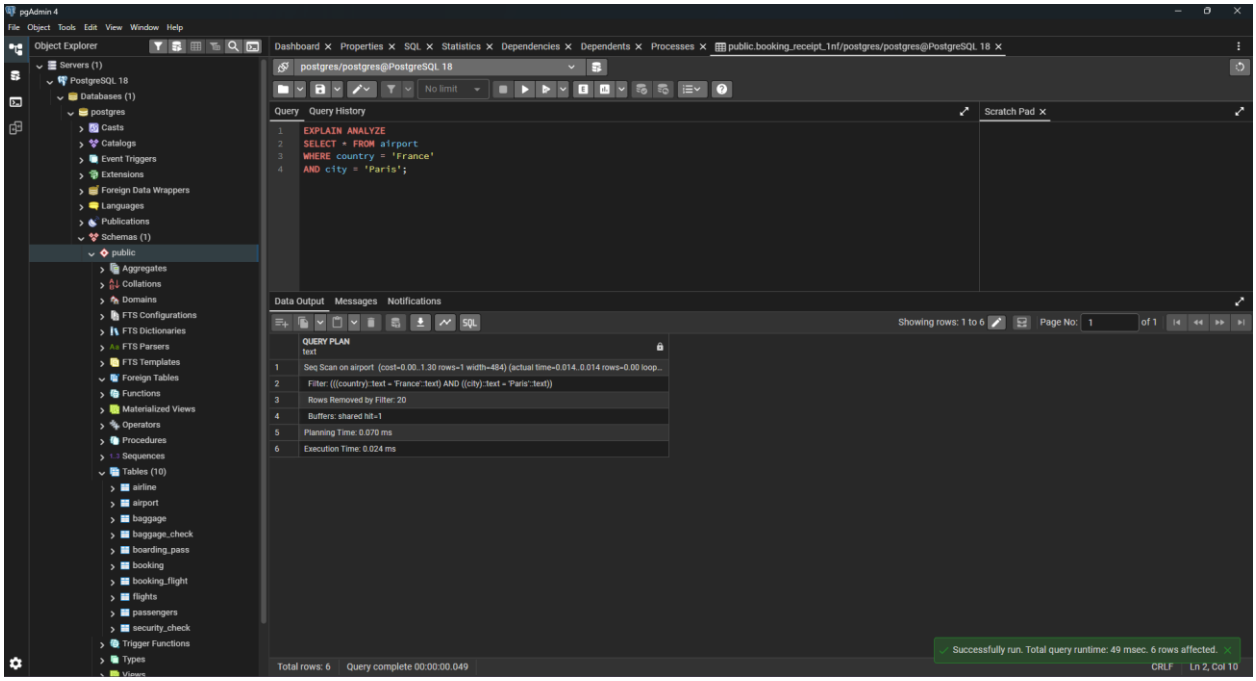


14. Create a composite index on the country and city columns of the airport table. Run three queries: filtering by country only, by city only, and by both columns. Check which of them use the index and explain the leftmost prefix rule using your results.

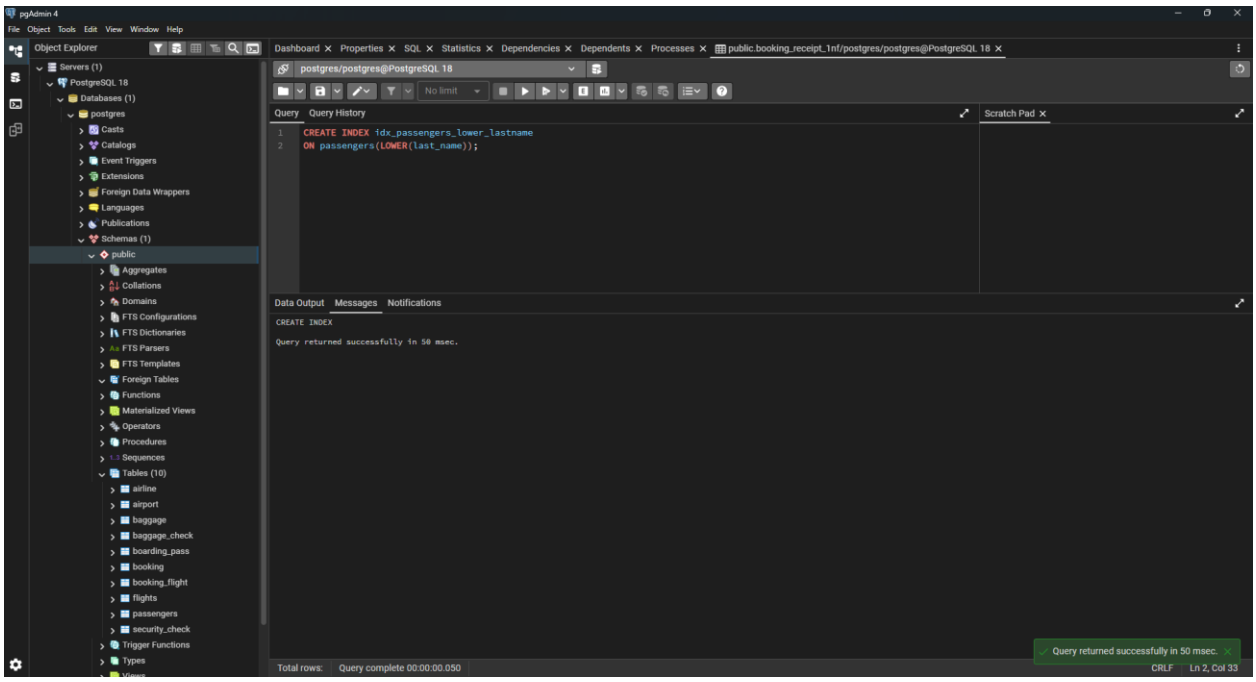

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public.booking_receipt_1nf/postgres/postgres@PostgreSQL 18
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1 CREATE INDEX idx_airport_country_city
2 ON airport(country, city);
Data Output
Messages
Notifications
CREATE INDEX
Query returned successfully in 58 msec.
Query returned successfully in 58 msec.
Total rows:
Query complete 00:00:00.058
CRLF
Ln 2, Col 27

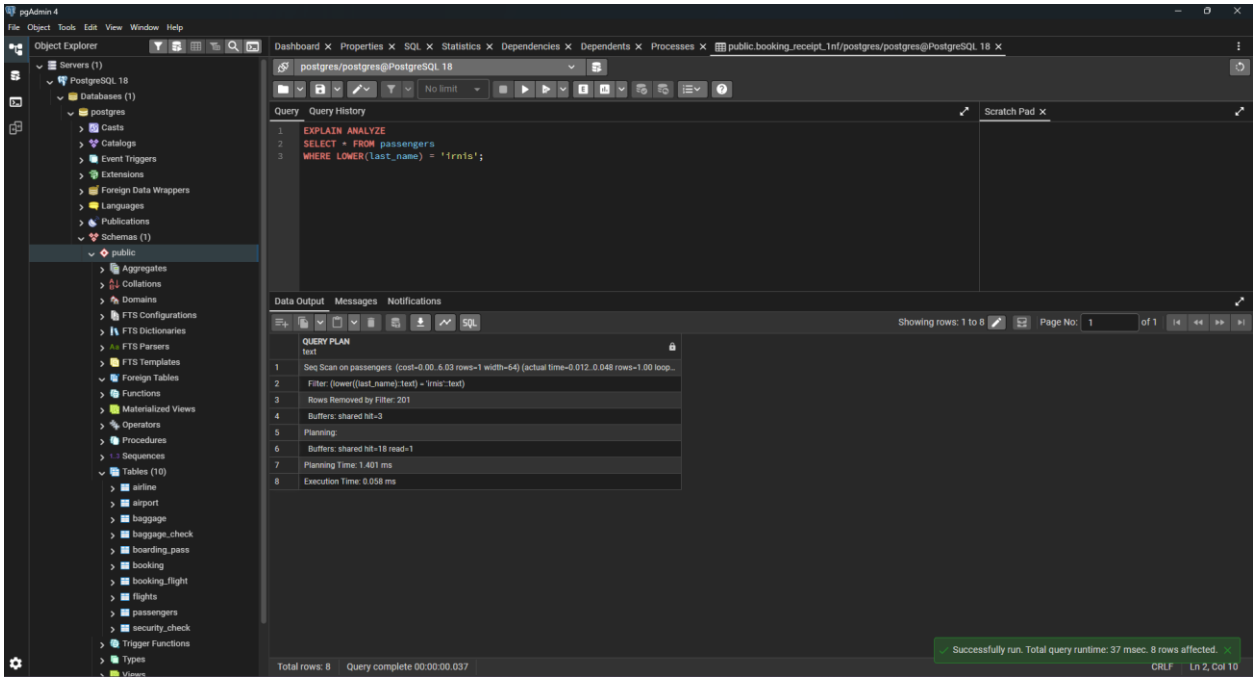

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postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1 EXPLAIN ANALYZE
2 SELECT *
3 FROM airport
4 WHERE country = 'France';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 8
Page No: 1 of 1
QUERY PLAN
text
1 Seq Scan on airport (cost=0.00..1.25 rows=1 width=484) (actual time=0.011..0.012 rows=0.00 loop...
2 Filter: ((country)::text = 'France'::text)
3 Rows Removed by Filter: 20
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=18 read=1
7 Planning Time: 1.496 ms
8 Execution Time: 0.022 ms
Successfully run. Total query runtime: 53 msec. 8 rows affected.
Total rows: 8
Query complete 00:00:00.053
CRLF
Ln 4, Col 26

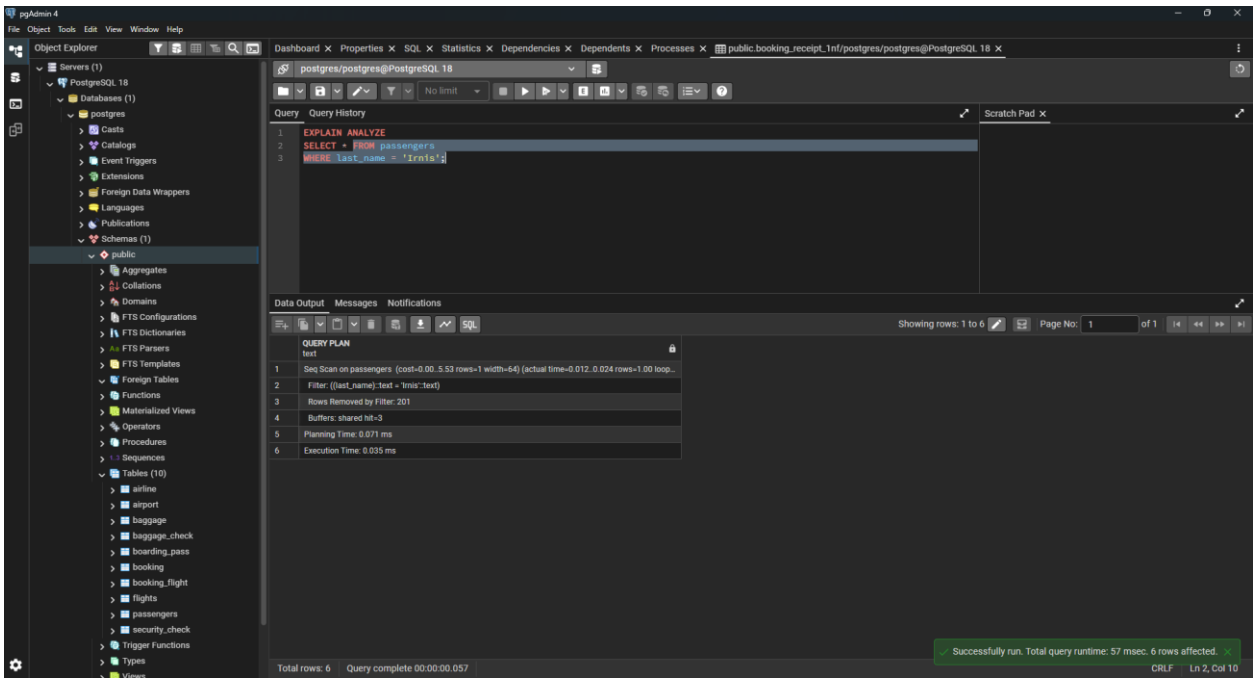




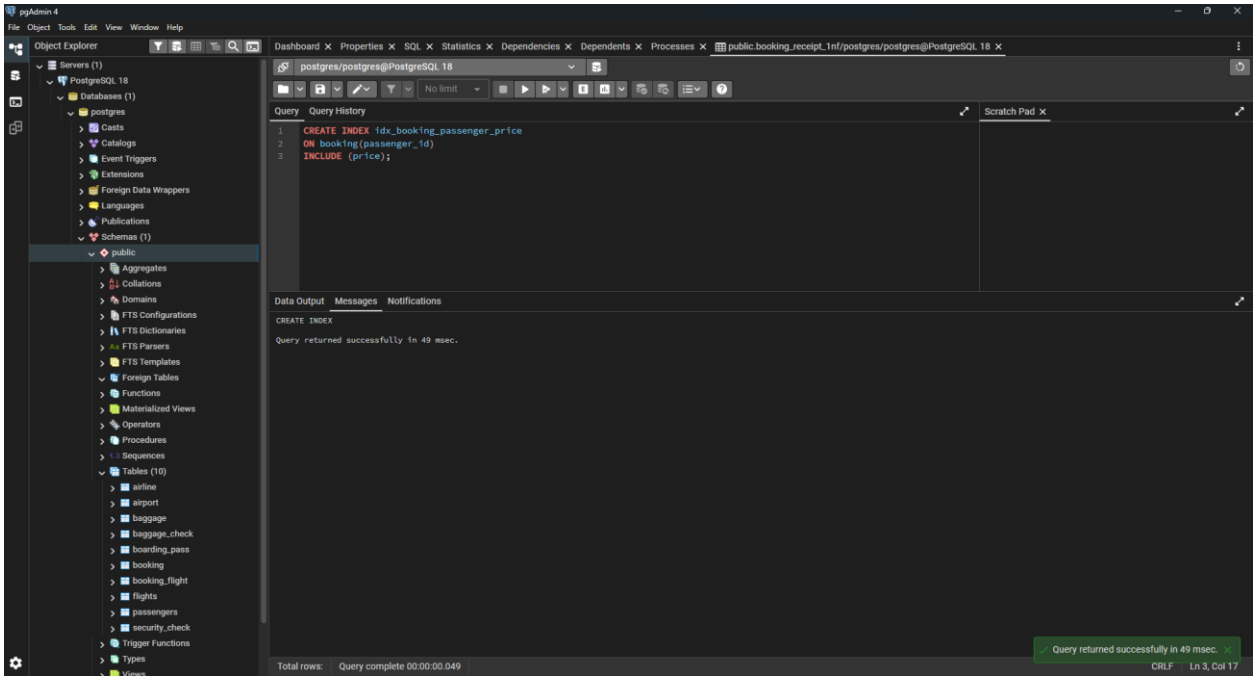
15. Create an expression (functional) index on the lower-case last_name of the Passengers table (LOWER(last_name)). Compare the plans of a query written as WHERE LOWER(last_name) = 'irnis' and a query written as WHERE last_name = 'Irnis'. Explain which query can use the index and why.

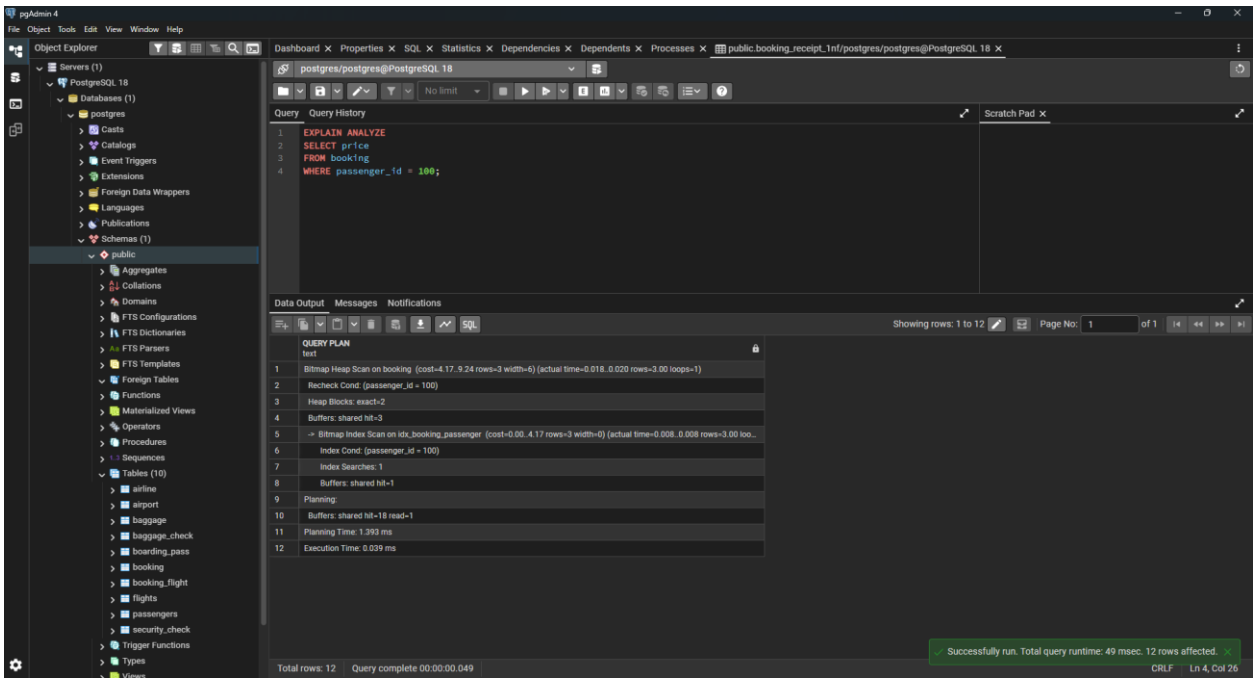

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public.booking_receipt_1nf/postgres/postgres@PostgreSQL 18
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1 CREATE INDEX idx_passengers_lower_lastname
2 ON passengers(LOWER(last_name));
Data Output
Messages
Notifications
CREATE INDEX
Query returned successfully in 50 msec.
Query returned successfully in 50 msec.
Total rows:
Query complete 00:00:00.050
CRLF
Ln 2, Col 33


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public.booking_receipt_1nf/postgres/postgres@PostgreSQL 18
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1 EXPLAIN ANALYZE
2 SELECT * FROM passengers
3 WHERE LOWER(last_name) = 'irnis';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 8
Page No: 1 of 1
QUERY PLAN
text
1 Seq Scan on passengers (cost=0.00..6.03 rows=1 width=64) (actual time=0.012..0.048 rows=1.00 loop...
2 Filter: (lower((last_name)::text) = 'irnis'::text)
3 Rows Removed by Filter: 201
4 Buffers: shared hit=3
5 Planning:
6 Buffers: shared hit=18 read=1
7 Planning Time: 1.401 ms
8 Execution Time: 0.058 ms
Successfully run. Total query runtime: 37 msec. 8 rows affected.
Total rows: 8
Query complete 00:00:00.037
CRLF
Ln 2, Col 10

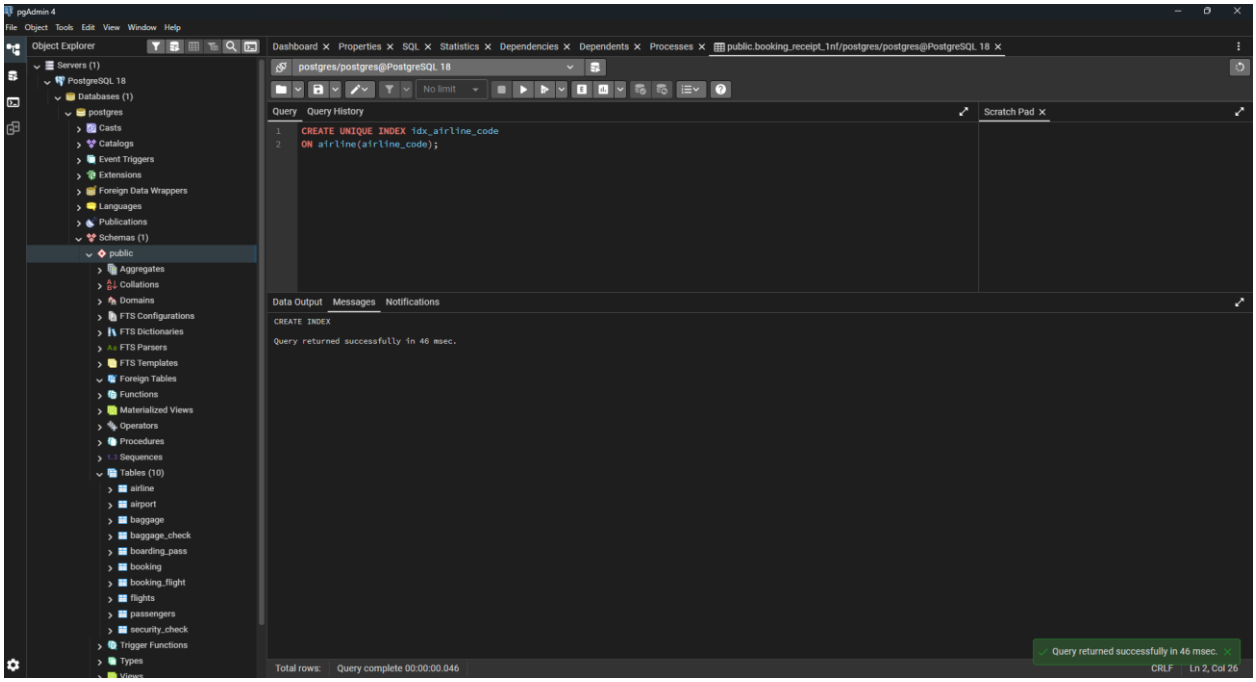


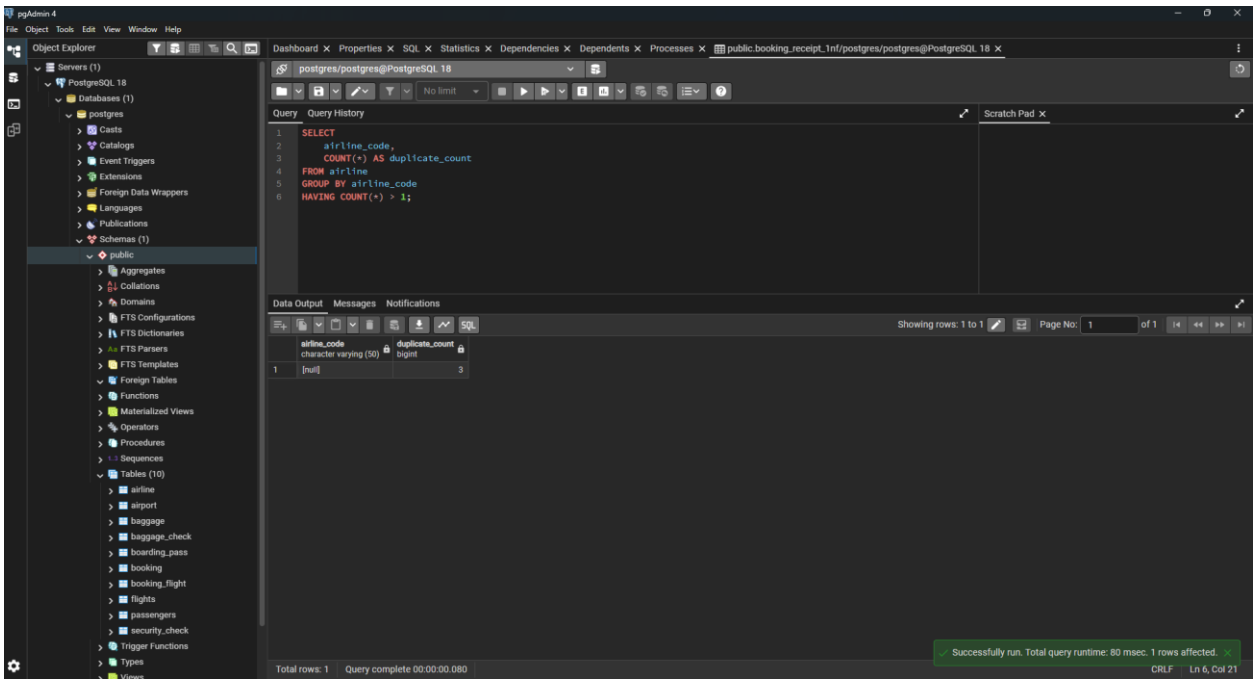
16. Create a covering index on the booking table for the column passenger_id and price. Write a query that returns only price for a given passenger_id and check with EXPLAIN ANALYZE whether an index-only scan (or "Using index") is used. Explain the benefits of a covering index.

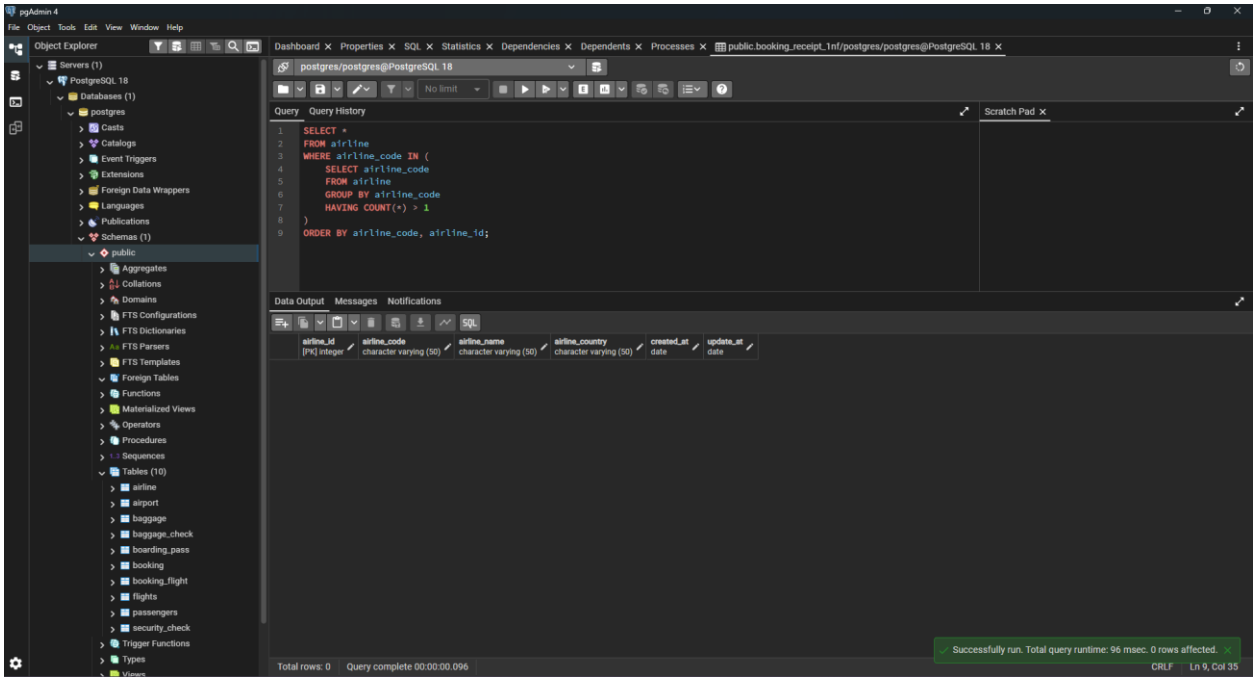




17. Try to create a unique index on the airline_code column of the airline table. If an error occurs, explain what caused it and find the problematic rows with a GROUP BY ... HAVING COUNT(*) > 1 query. Then create the unique index in a way that works for the current data, or after fixing the data.




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No limit
Query Query History
Scratch Pad
SELECT
airline_code,
COUNT(*) AS duplicate_count
FROM airline
GROUP BY airline_code
HAVING COUNT(*) > 1;
Data Output Messages Notifications
SQL
Showing rows: 1 to 1
Page No: 1 of 1
airline_code character varying (50)
duplicate_count bigint
1 [null] 3
Successfully run. Total query runtime: 80 msec. 1 rows affected.
Total rows: 1 Query complete 00:00:00.080
CRLF Ln 6, Col 21


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postgres/postgres@PostgreSQL 18
No limit
Query Query History
Scratch Pad
SELECT *
FROM airline
WHERE airline_code IN (
SELECT airline_code
FROM airline
GROUP BY airline_code
HAVING COUNT(*) > 1
)
ORDER BY airline_code, airline_id;
Data Output Messages Notifications
SQL
airline_id [PK] integer
airline_code character varying (50)
airline_name character varying (50)
airline_country character varying (50)
created_at date
update_at date
Successfully run. Total query runtime: 96 msec. 0 rows affected.
Total rows: 0 Query complete 00:00:00.096
CRLF Ln 9, Col 35